MOCCA 2025
Fundamentals of Character Animation and Motion Control

Lecture 09

Learning to Walk

Libin Liu

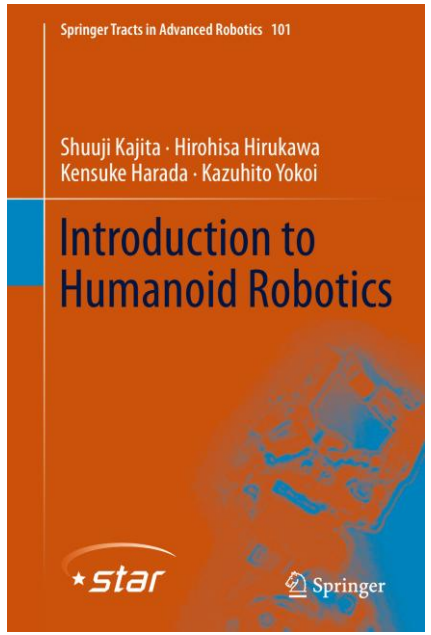
School of Intelligence Science and Technology
Peking University

VCL @ PKU

Outline

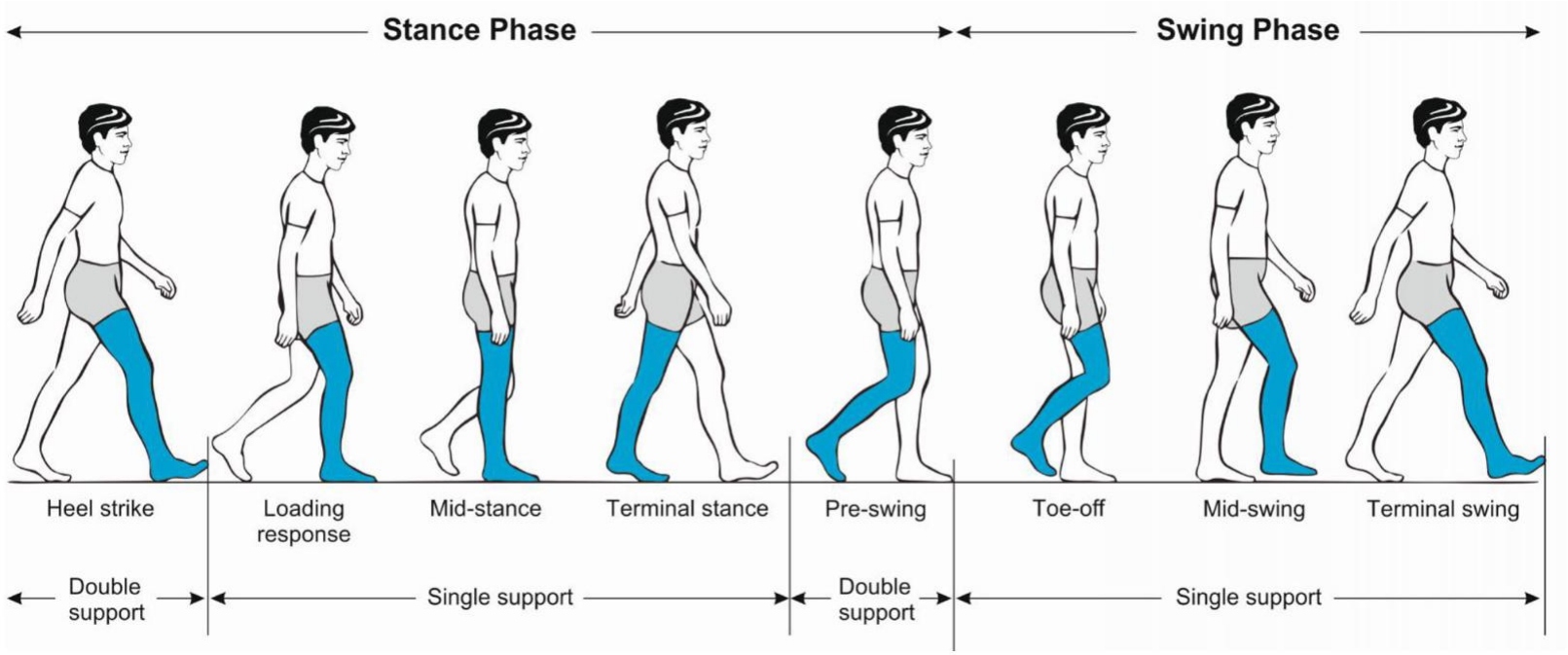
- Walking and Dynamic Balance

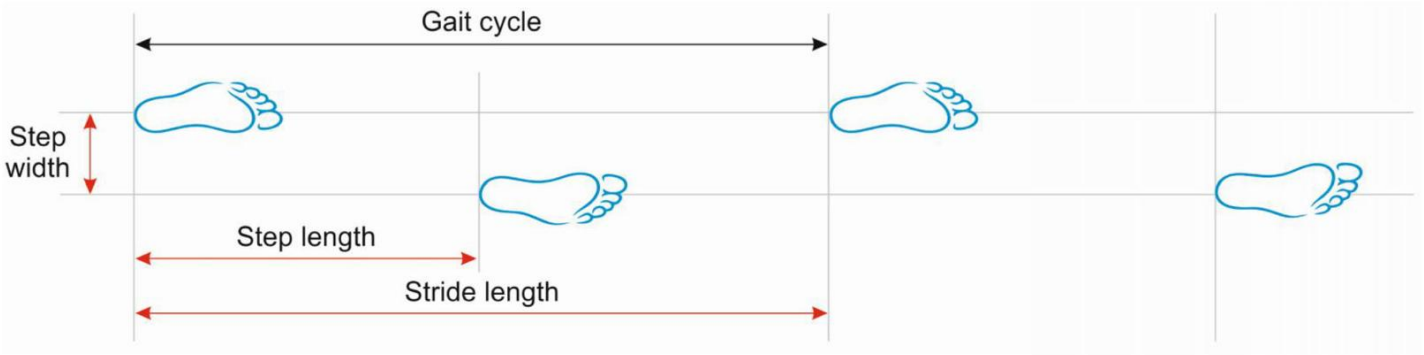
- Simplified Models
  - ZMP (Zero-Moment Point)
  - Inverted Pendulum
  - Cart-Table Model
  - SIMBICON



[Kajita et al. Introduction to Humanoid Robotics]

Walking

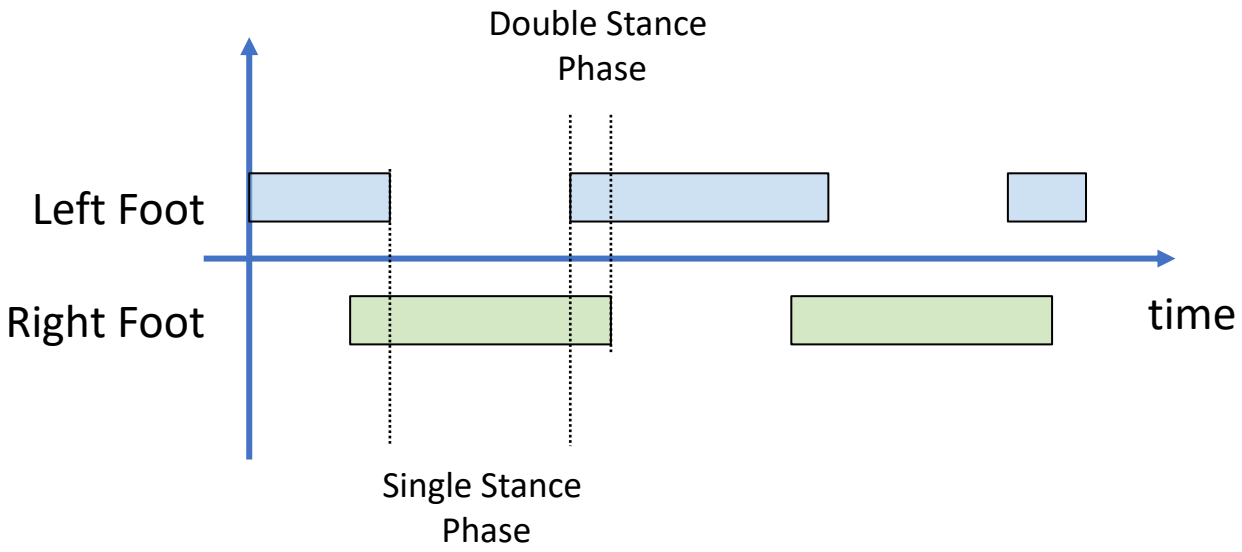




phases of a walking gait cycle

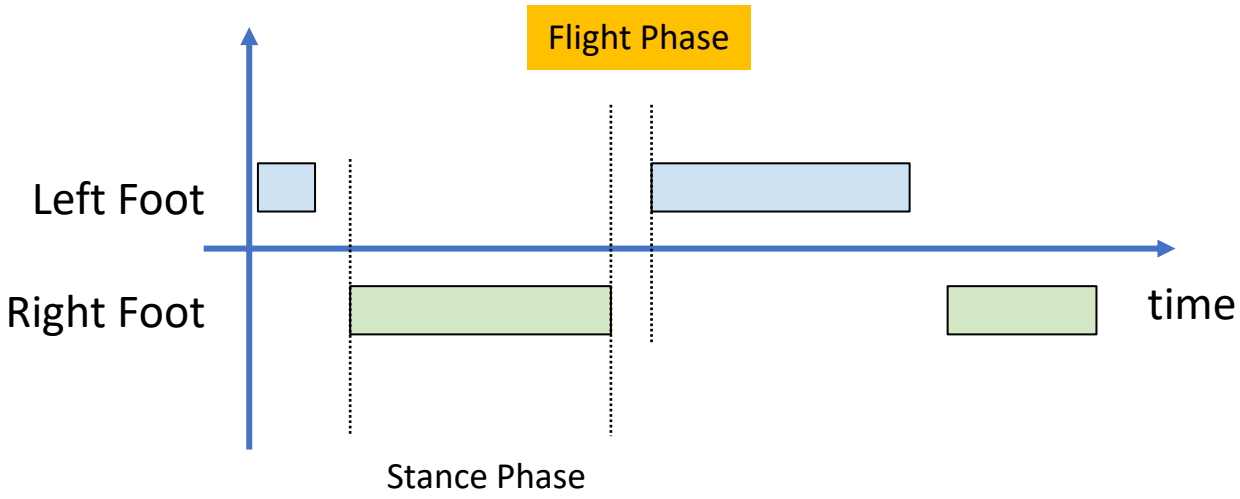
Pirker and Katzenschlager 2017.
Gait disorders in adults and the elderly.

Walking



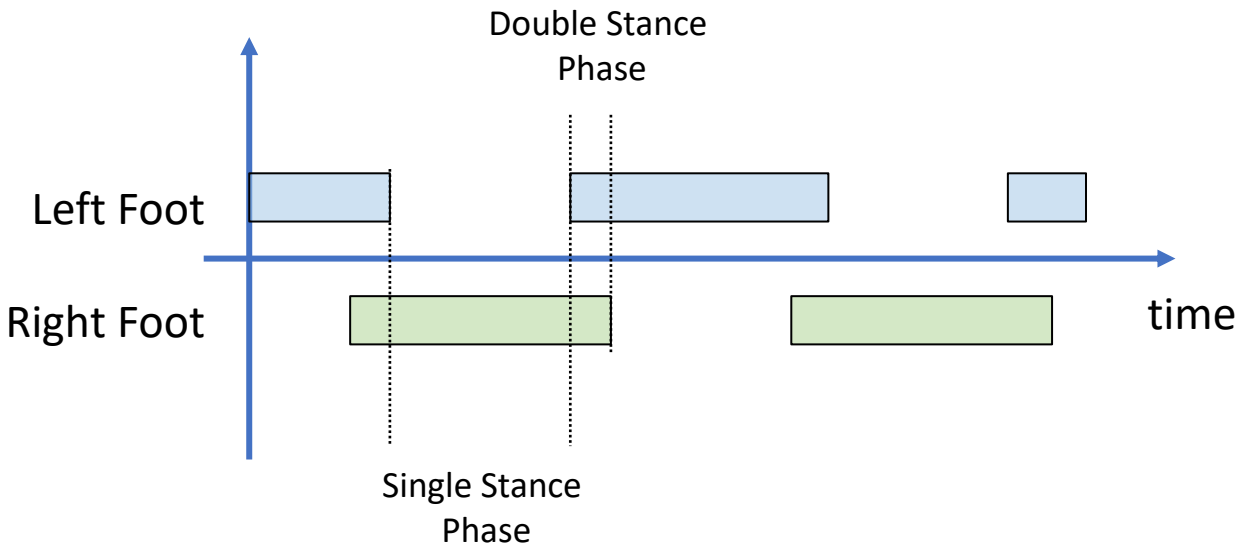
Walking: move without *loss of contact*, or flight phases

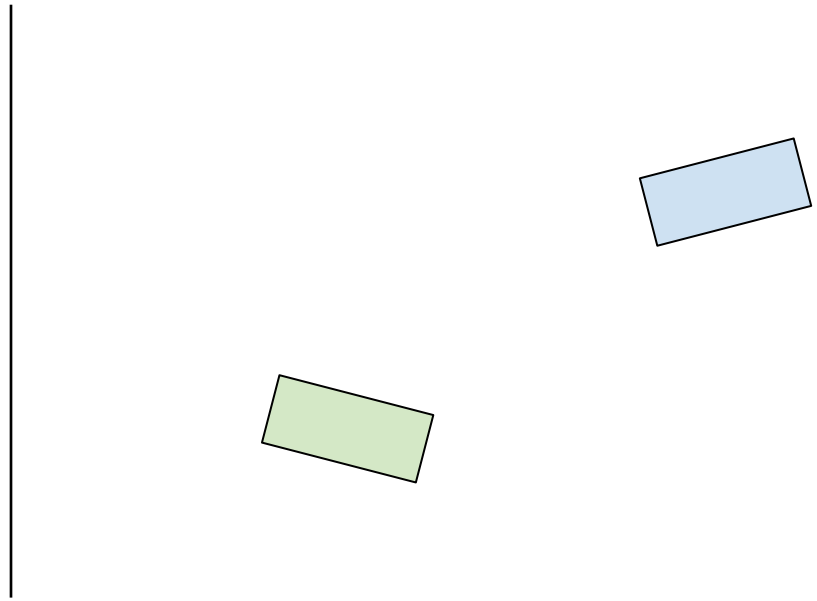
Walking



Running

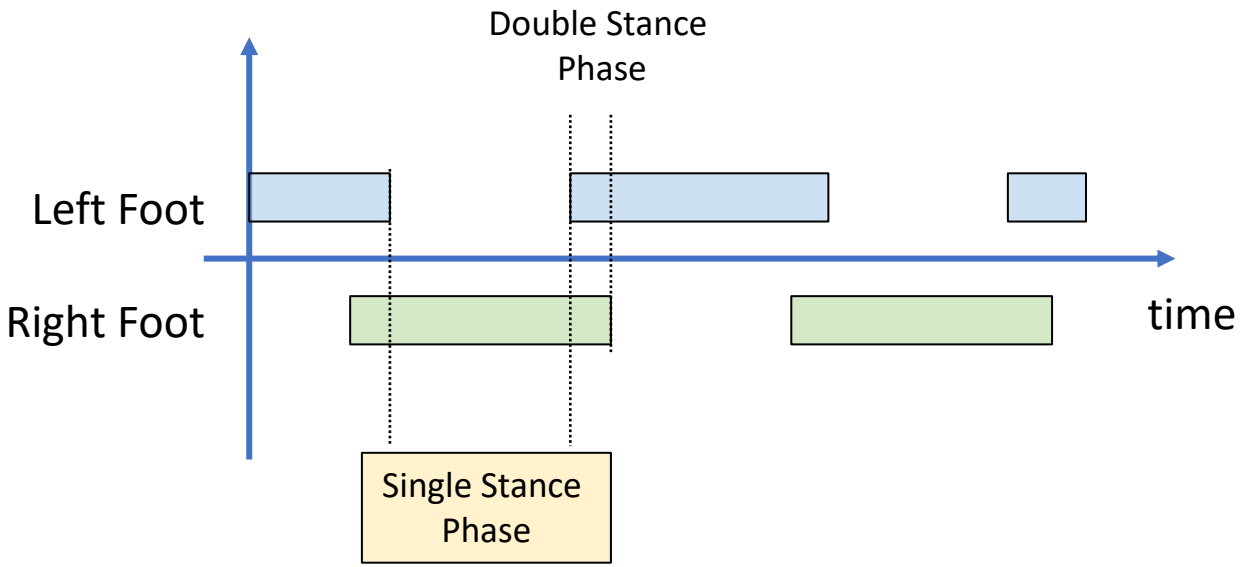
Walking with Static Balance

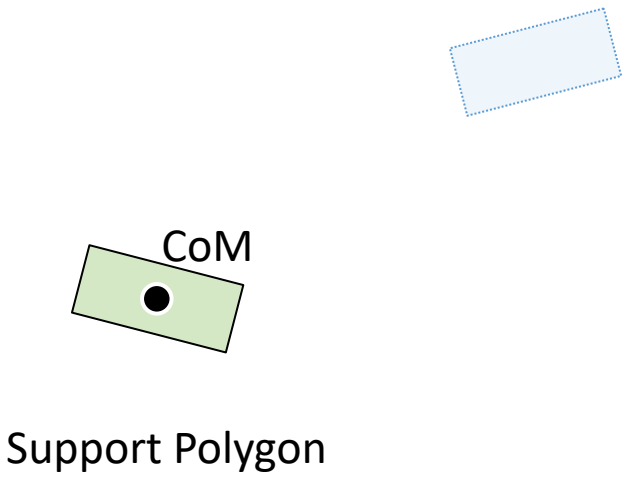




Walking: move without *loss of contact*, or flight phases

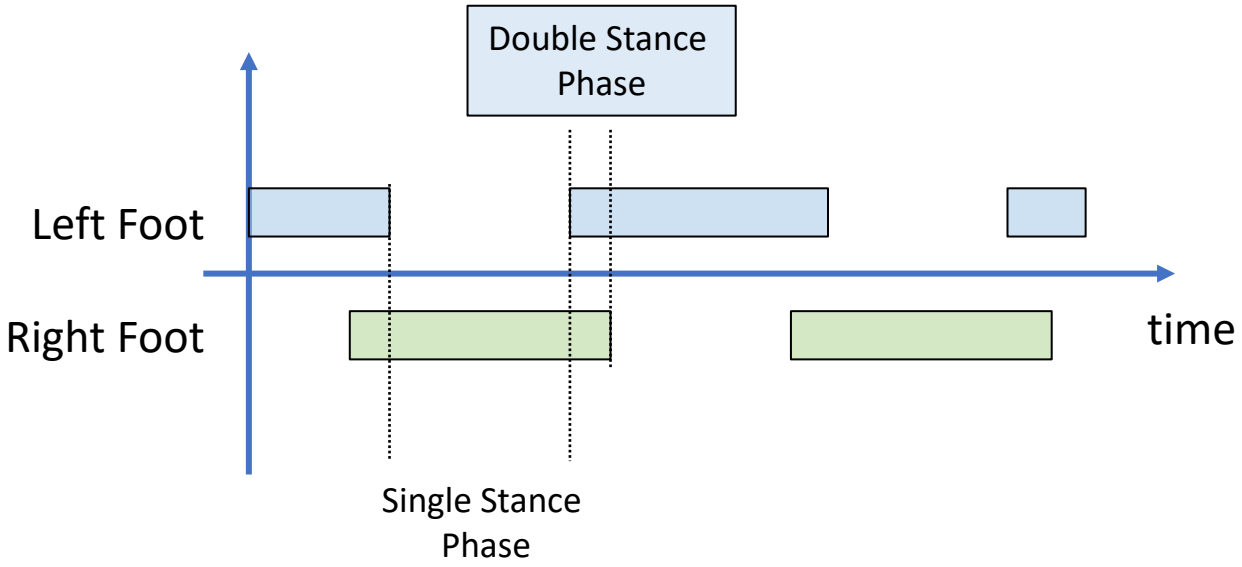
Walking with Static Balance

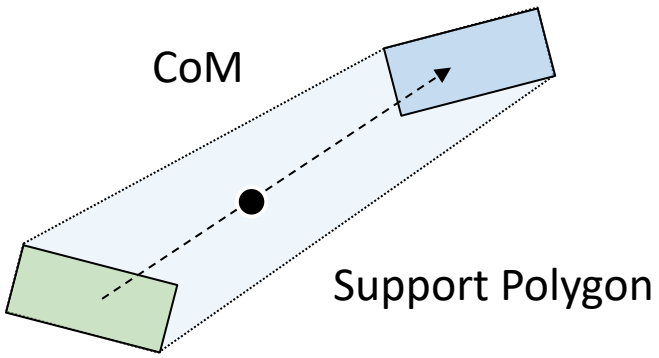




Walking: move without *loss of contact*, or flight phases

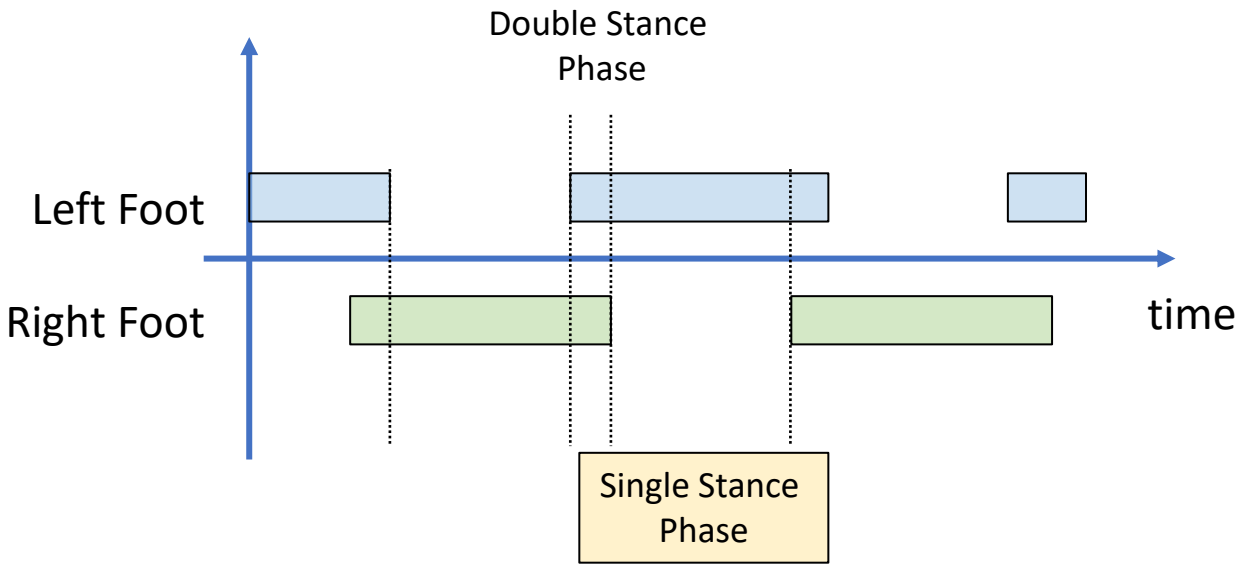
Walking with Static Balance

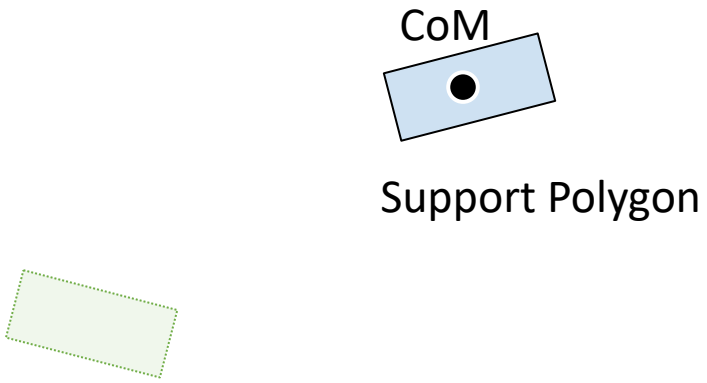




Walking: move without *loss of contact*, or flight phases

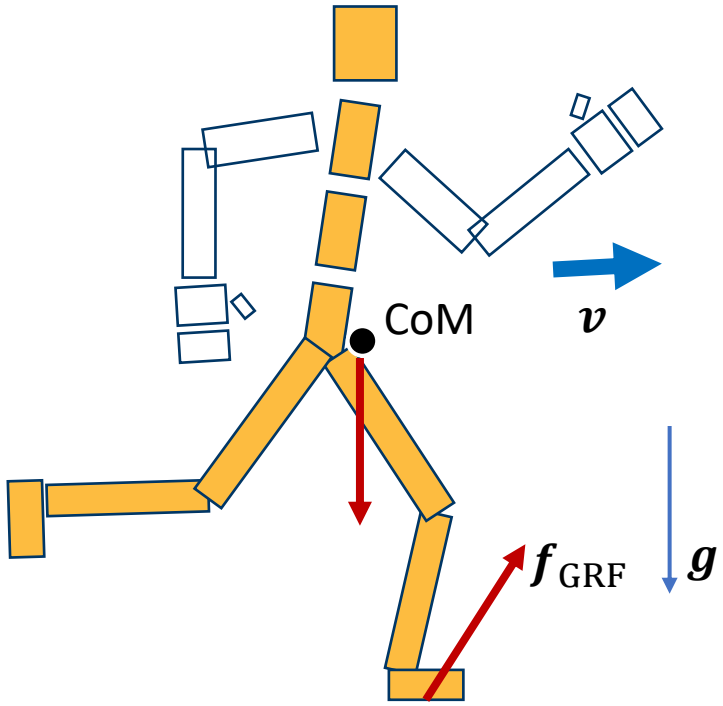
Walking with Static Balance





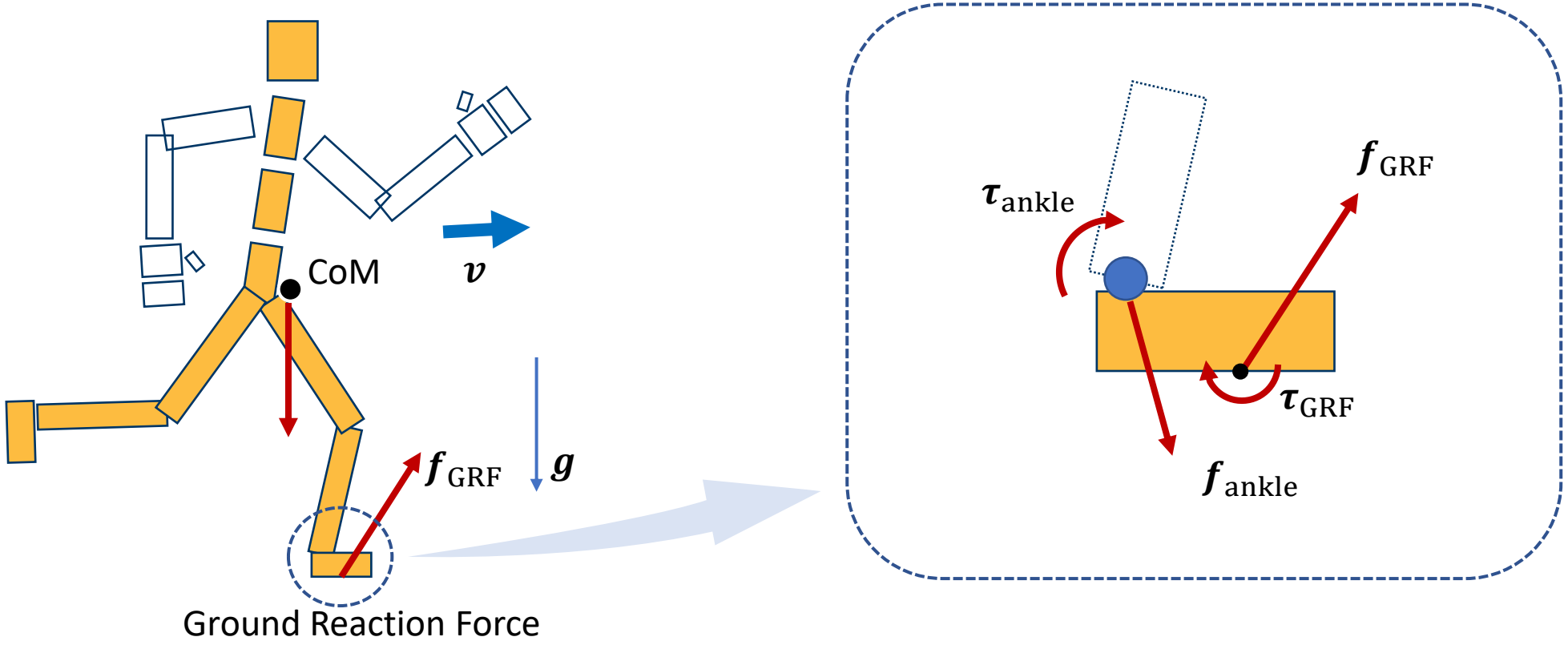
Walking: move without *loss of contact*, or flight phases

Zero-Moment Point (ZMP)

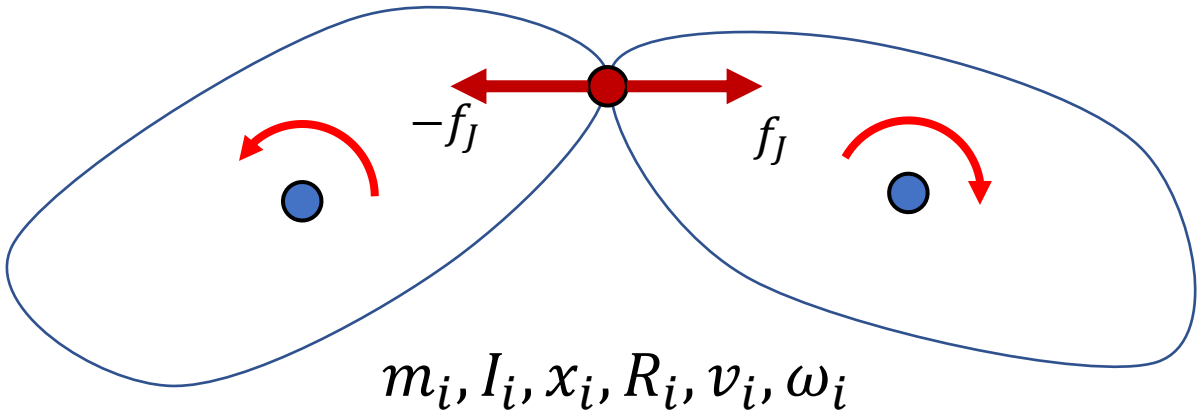


Ground Reaction Force

Zero-Moment Point (ZMP)

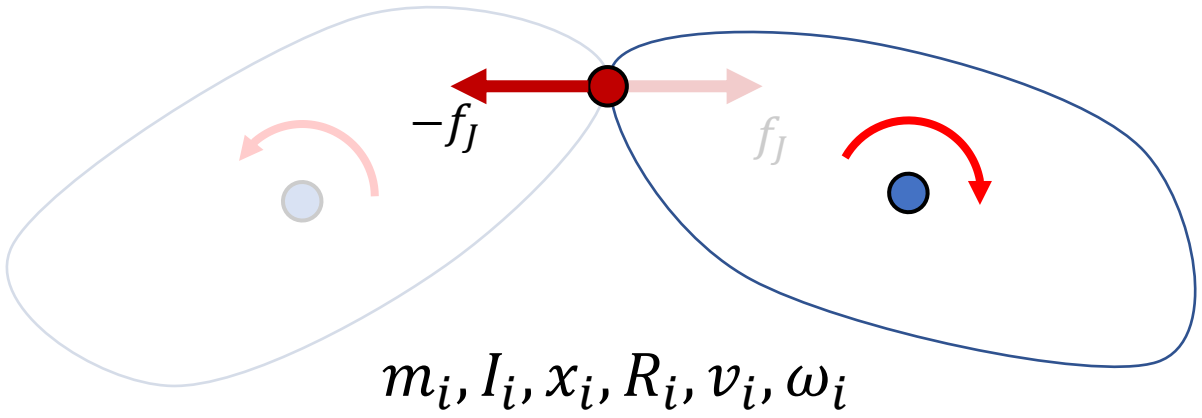


Recall: A System of Links and Joints



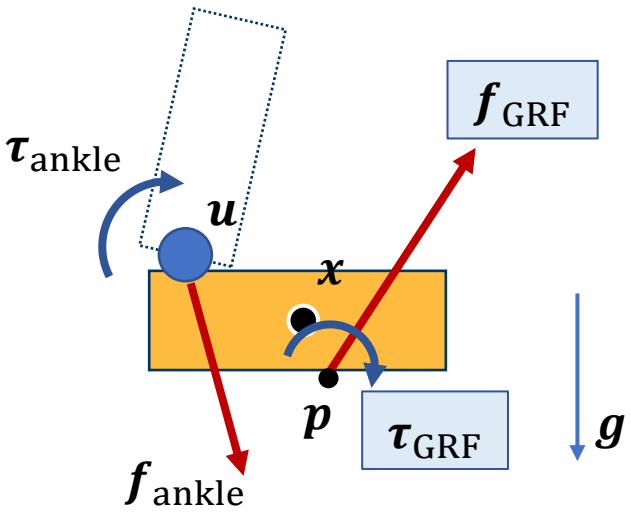
$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{f}_J$$

Recall: A System of Links and Joints



$$M\dot{\boldsymbol{v}} + C(\boldsymbol{x}, \boldsymbol{v}) = \boldsymbol{f} + \boldsymbol{f}_J$$

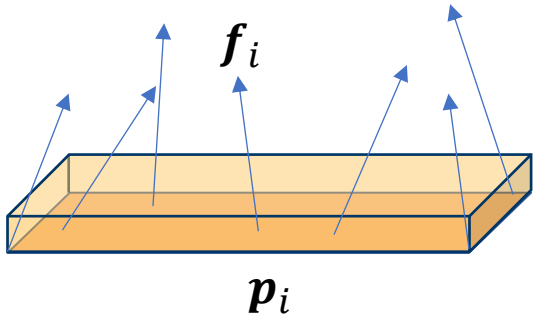
Zero-Moment Point (ZMP)



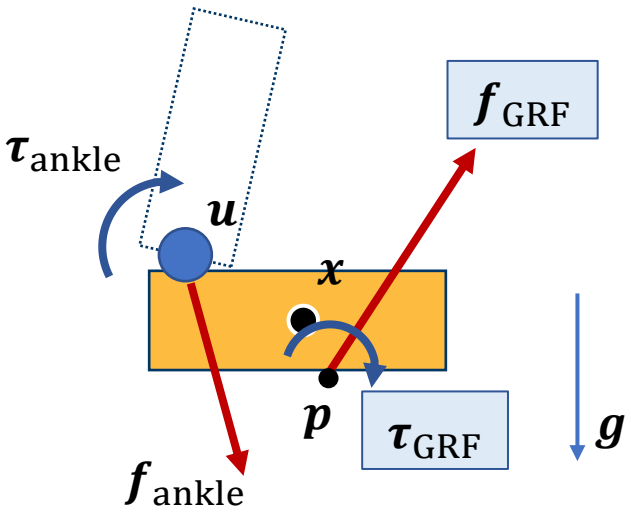
normal force

$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$

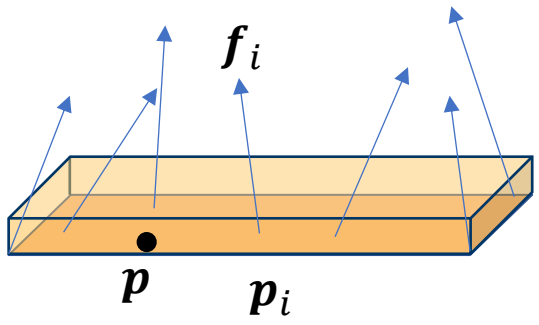
friction



Zero-Moment Point (ZMP)



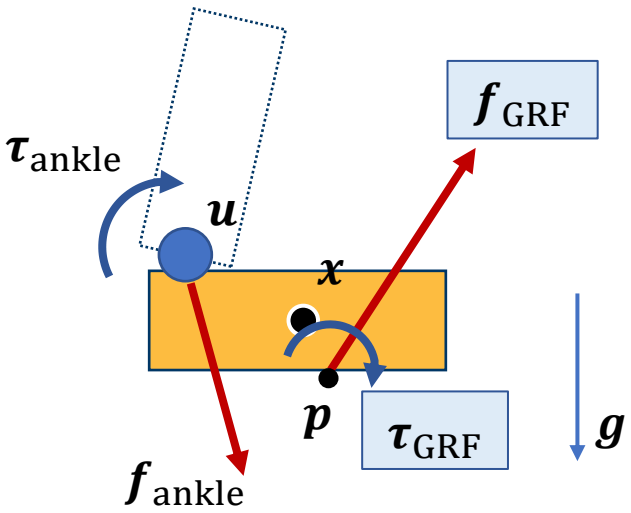
$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



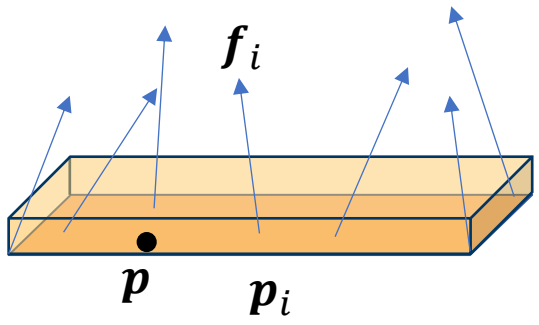
$$\boldsymbol{f}_{\text{GRF}} = \sum_i \boldsymbol{f}_i$$

$$\boldsymbol{\tau}_{\text{GRF}} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i$$

Zero-Moment Point (ZMP)



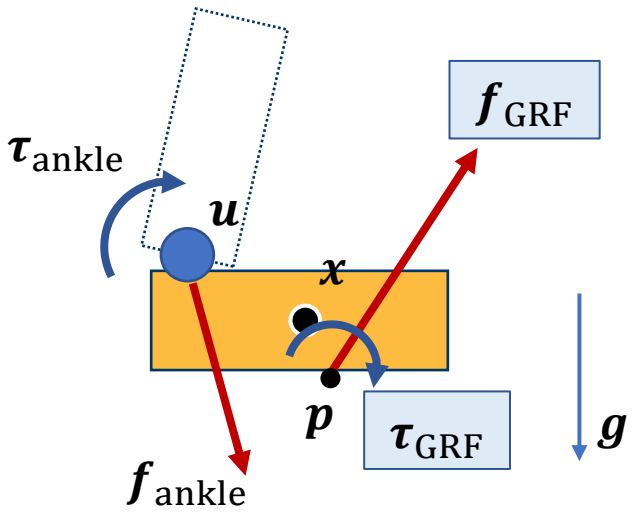
$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



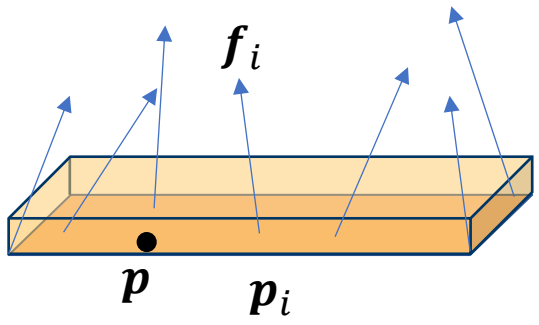
$$\boldsymbol{\tau}_{\text{GRF}} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i$$

Assuming the ground is flat and level
so $\boldsymbol{p}_i - \boldsymbol{p}$ is always in the horizontal plane

Zero-Moment Point (ZMP)



$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



$$\boldsymbol{\tau}_{\text{GRF}} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i$$

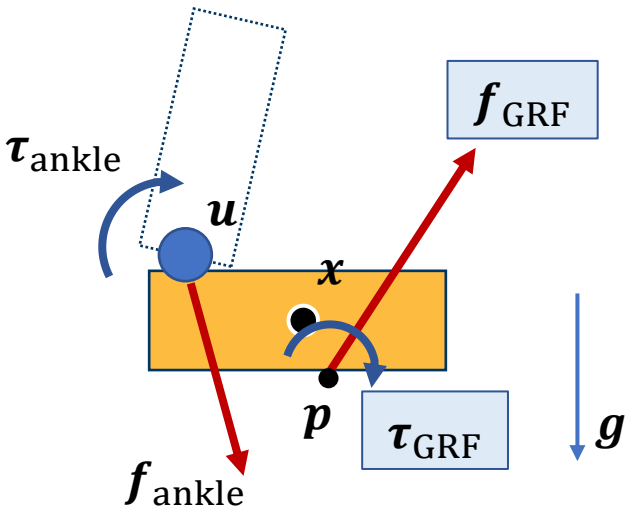
Assuming the ground is flat and level
so $\boldsymbol{p}_i - \boldsymbol{p}$ is always in the horizontal plane

$$\boldsymbol{\tau}_{\text{GRF}} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \left(f_i^y + f_i^{xz}\right)$$

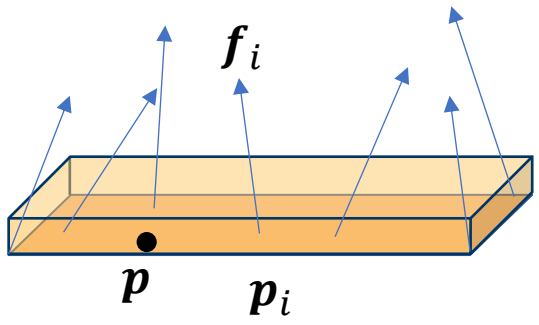
horizontal vertical

$$= \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times f_i^y + \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times f_i^{xz}$$

Zero-Moment Point (ZMP)



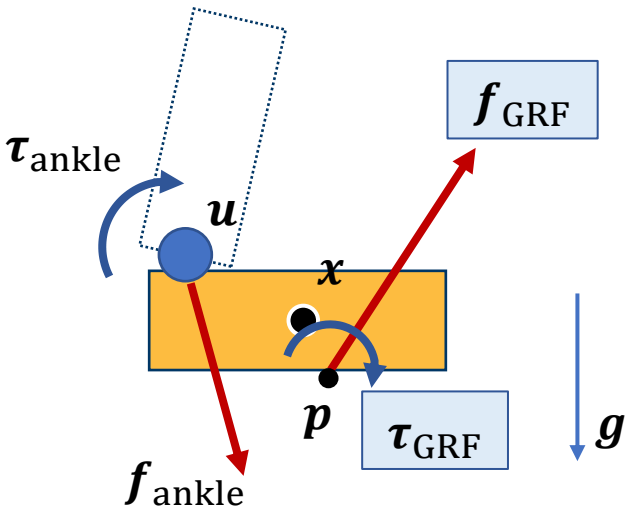
$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



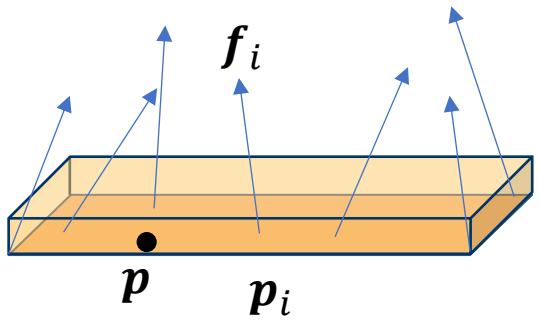
$$\boldsymbol{\tau}_{\text{GRF}}^y = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^{xz}$$

$$\boldsymbol{\tau}_{\text{GRF}}^{xz} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^y$$

Zero-Moment Point (ZMP)



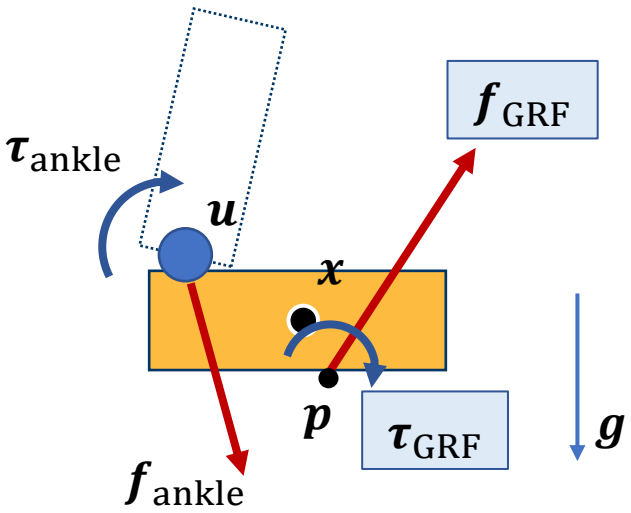
$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



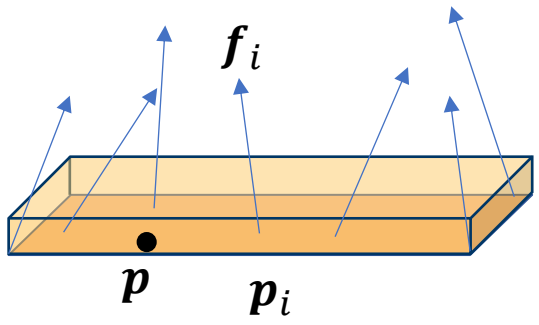
$$\boldsymbol{\tau}_{\text{GRF}}^{xz} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^y$$

$$= \sum_i \boldsymbol{p}_i \times \boldsymbol{f}_i^y - \boldsymbol{p} \times \left(\sum_i f_i^y \right) \boldsymbol{y}$$

Zero-Moment Point (ZMP)



$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$

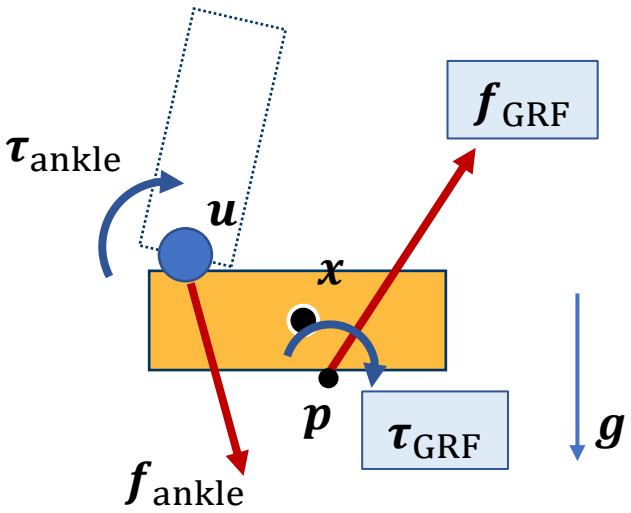


$$\boldsymbol{\tau}_{\text{GRF}}^{xz} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^y$$

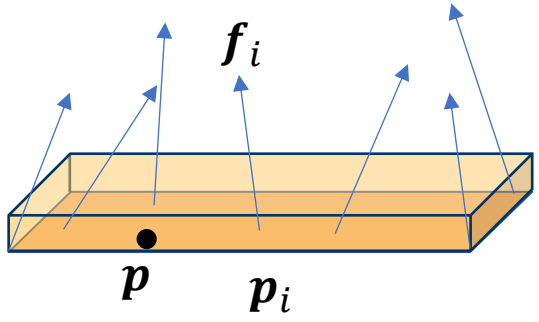
$$= \sum_i \boldsymbol{p}_i \times \boldsymbol{f}_i^y - \boldsymbol{p} \times \left(\sum_i f_i^y \right) \boldsymbol{y}$$

Can we find $\boldsymbol{p}$ such that $\boldsymbol{\tau}_{\text{GRF}}^{xz} = 0$?

Zero-Moment Point (ZMP)



$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$

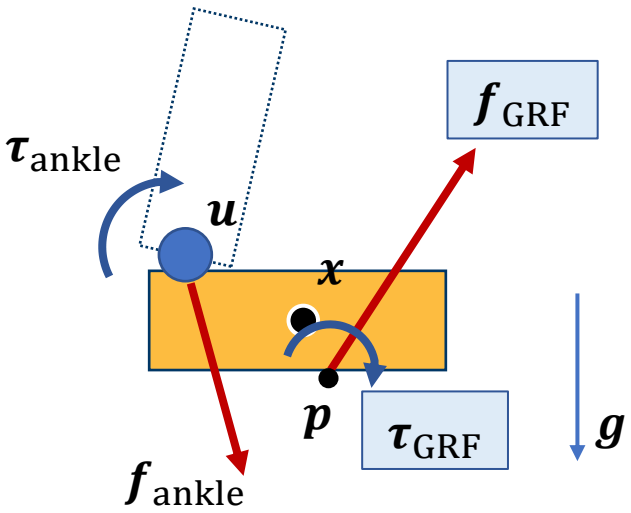


$$\boldsymbol{\tau}_{\text{GRF}}^{xz} = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^y$$

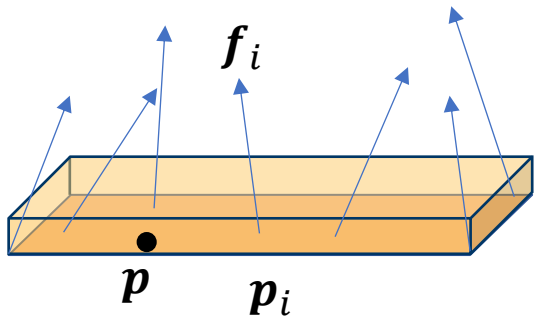
$$= \sum_i \boldsymbol{p}_i \times \boldsymbol{f}_i^y - \boldsymbol{p} \times \left(\sum_i f_i^y \right) \boldsymbol{y}$$

Center of Pressure $\boldsymbol{p} = \dfrac{\sum_i \boldsymbol{p}_i f_i^y}{\sum_i f_i^y}$ $\Rightarrow$ $\boldsymbol{\tau}_{\text{GRF}}^{xz} = 0$

Zero-Moment Point (ZMP)



$$\boldsymbol{f}_i = \boldsymbol{f}_i^y + \boldsymbol{f}_i^{xz}$$



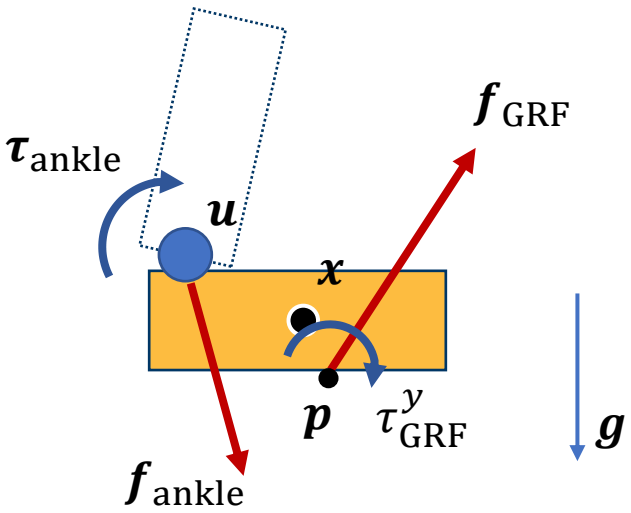
Center of Pressure

$$\boldsymbol{p} = \frac{\sum_i \boldsymbol{p}_i f_i^y}{\sum_i f_i^y}$$

$$\boldsymbol{f}_{\text{GRF}} = \sum_i \boldsymbol{f}_i$$

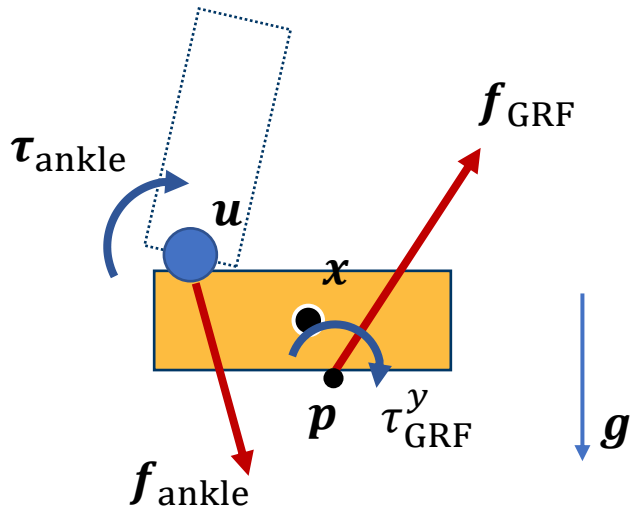
$$\boldsymbol{\tau}_{\text{GRF}} = \tau_{\text{GRF}}^y = \sum_i (\boldsymbol{p}_i - \boldsymbol{p}) \times \boldsymbol{f}_i^{xz}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

Zero-Moment Point (ZMP)



The position of $\boldsymbol{p}$ is not known, but we assume

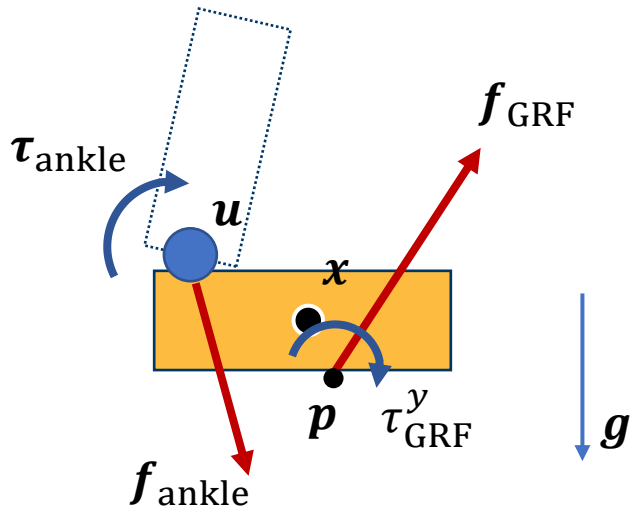
$$\tau^{xz}_{\text{GRF}} = \mathbf{0}$$

So

$$\boldsymbol{\tau}_{\text{GRF}} = \tau^{y}_{\text{GRF}}$$

The foot should not move in a **stance phase**

Zero-Moment Point (ZMP)

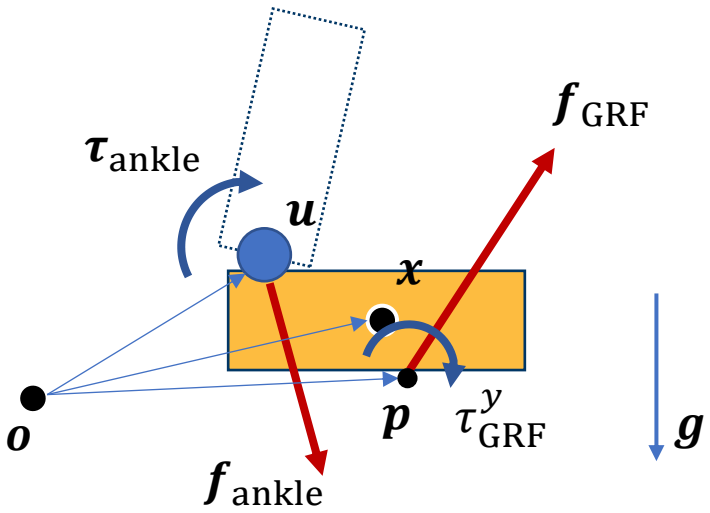


Static Equilibrium:

$$\boldsymbol{f}_{\text{ankle}} + \boldsymbol{f}_{\text{GRF}} + m\boldsymbol{g} = \boldsymbol{0}$$

The foot should not move in a **stance phase**

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

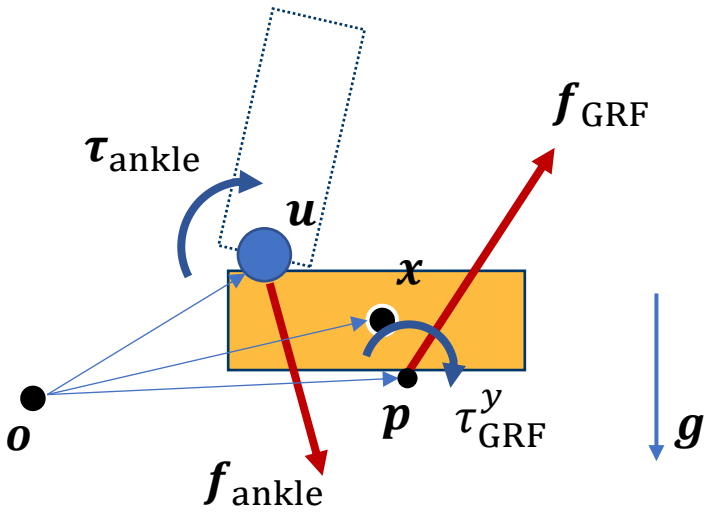
Static Equilibrium:

$$\boldsymbol{f}_{\text{ankle}} + \boldsymbol{f}_{\text{GRF}} + m\boldsymbol{g} = \boldsymbol{0}$$

The moment around a reference point $\boldsymbol{o}$:

$$(\boldsymbol{u} - \boldsymbol{o}) \times \boldsymbol{f}_{\text{ankle}} + (\boldsymbol{p} - \boldsymbol{o}) \times \boldsymbol{f}_{\text{GRF}} + (\boldsymbol{x} - \boldsymbol{o}) \times m\boldsymbol{g}$$
$$+\tau^{y}_{\text{GRF}} + \boldsymbol{\tau}_{\text{ankle}} = \boldsymbol{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

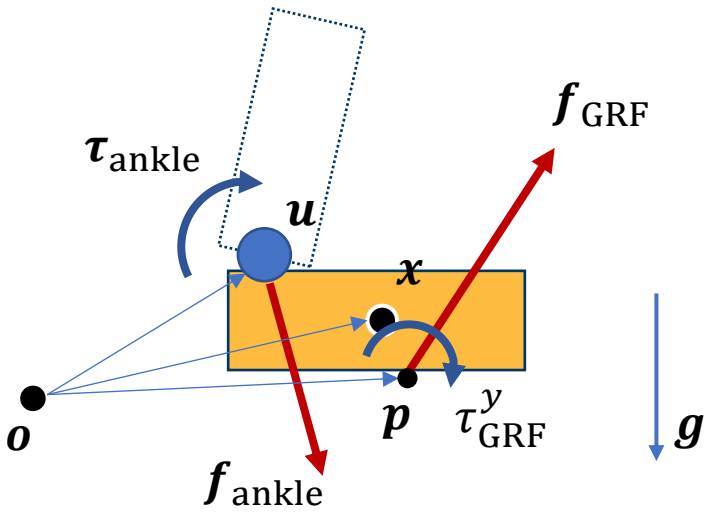
The moment around a reference point $\boldsymbol{o}$:

$$(\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}+(\boldsymbol{p}-\boldsymbol{o})\times\boldsymbol{f}_{\text{GRF}}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}$$
$$+\tau^{y}_{\text{GRF}}+\boldsymbol{\tau}_{\text{ankle}}=\boldsymbol{0}$$

Horizontal components (moment projected onto xz plane):

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}$$
$$+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\boldsymbol{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

The moment around a reference point $\boldsymbol{o}$:

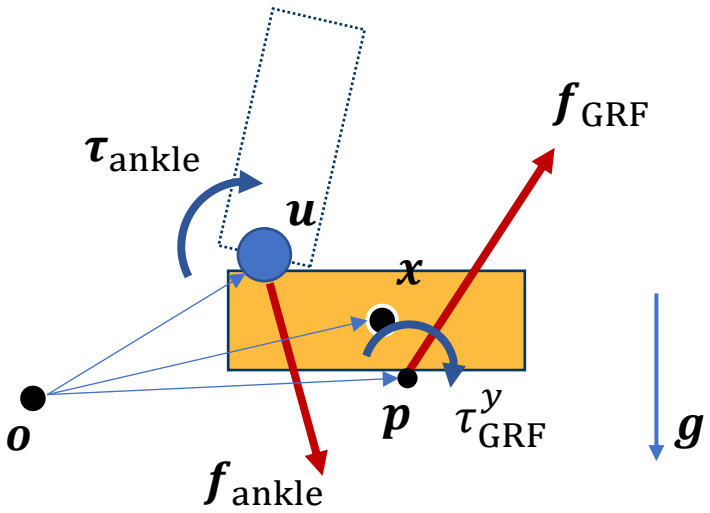
$$(\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}+(\boldsymbol{p}-\boldsymbol{o})\times\boldsymbol{f}_{\text{GRF}}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}$$
$$+\tau^{y}_{\text{GRF}}+\boldsymbol{\tau}_{\text{ankle}}=\boldsymbol{0}$$

Horizontal components (moment projected onto xz plane):

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}$$
$$+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\boldsymbol{0}$$

We can solve this equation to find $\boldsymbol{p}$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

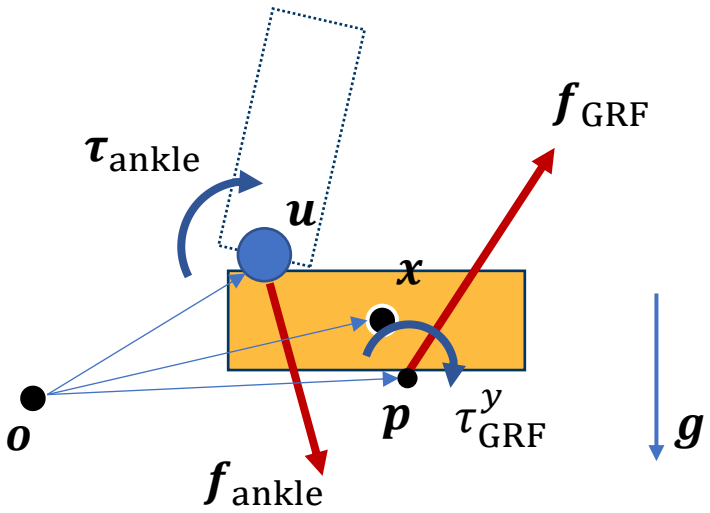
$\boldsymbol{p}$ is called Zero-Moment Point (ZMP) because it makes

$$\tau^{xz}_{\text{GRF}} = \mathbf{0}$$

and the horizontal moment

$$\left((\boldsymbol{u}-\boldsymbol{o})\times \boldsymbol{f}_{\text{ankle}}\right)^{xz} + \left((\boldsymbol{p}-\mathbf{o})\times \boldsymbol{f}_{\text{GRF}}\right)^{xz} + (\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g} + \boldsymbol{\tau}^{xz}_{\text{ankle}} = \mathbf{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

$\boldsymbol{p}$ is called Zero-Moment Point (ZMP) because it makes

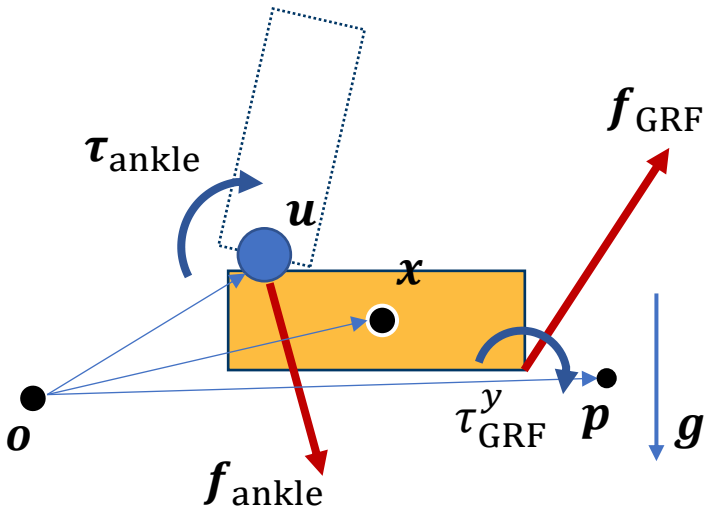
$$\tau^{xz}_{\text{GRF}} = \mathbf{0}$$

and the horizontal moment

$$\left((\boldsymbol{u}-\boldsymbol{o})\times \boldsymbol{f}_{\text{ankle}}\right)^{xz} + \left((\boldsymbol{p}-\mathbf{o})\times \boldsymbol{f}_{\text{GRF}}\right)^{xz} + (\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g} + \boldsymbol{\tau}^{xz}_{\text{ankle}} = \mathbf{0}$$

Only when p is within the support polygon!

Zero-Moment Point (ZMP)



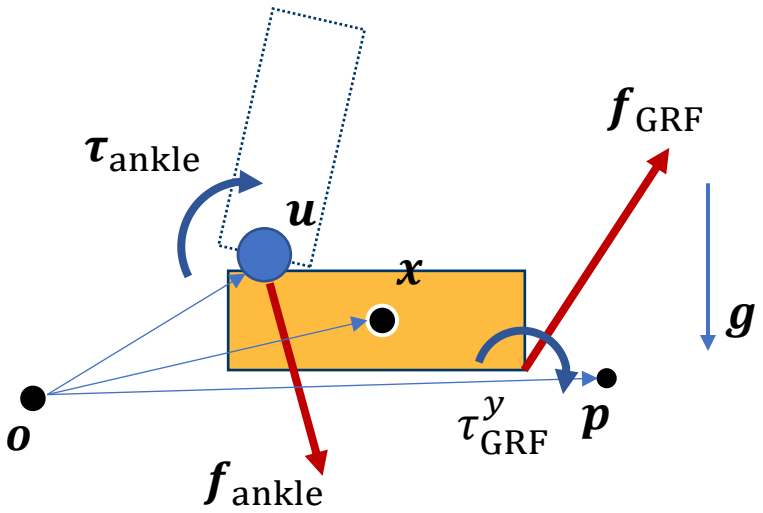
The foot should not move in a **stance phase**

If the solution of

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}$$
$$+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\mathbf{0}$$

$\boldsymbol{p}$ is outside the support polygon

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

If the solution of

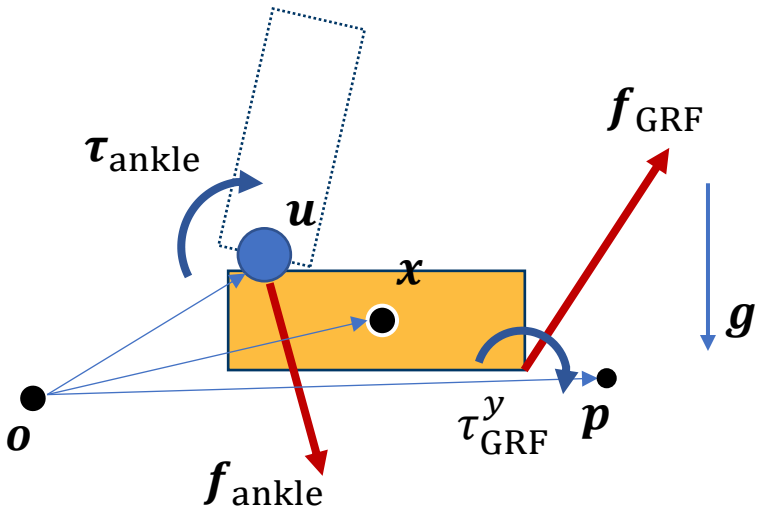
$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\mathbf{0}$$

$\boldsymbol{p}$ is outside the support polygon

$\boldsymbol{p}$ could NOT be the center of pressure, because all the GRFs are applied within the polygon, so that

$$\tau^{xz}_{\text{GRF}}\neq\mathbf{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

If the solution of

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\mathbf{0}$$

$\boldsymbol{p}$ is outside the support polygon

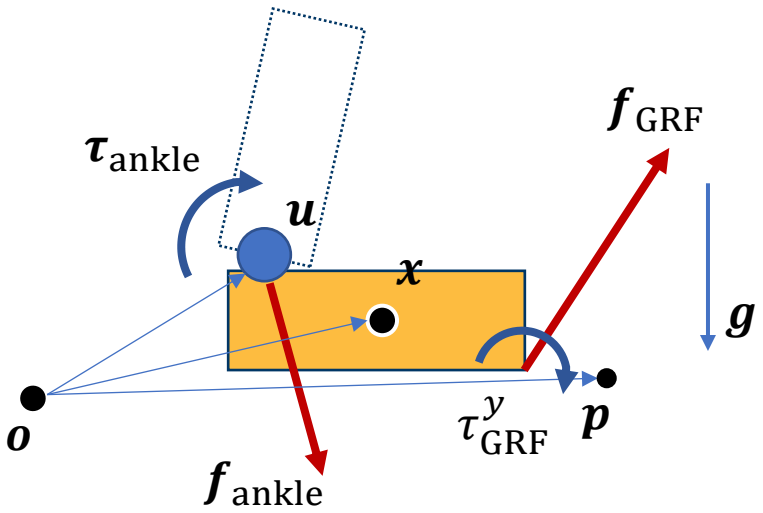
$\boldsymbol{p}$ could NOT be the center of pressure, because all the GRFs are applied within the polygon, so that

$$\tau^{xz}_{\text{GRF}}\neq\mathbf{0}$$

Or, if $\boldsymbol{p}'$ is the real center of pressure, we have

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}'-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}\neq\mathbf{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

If the solution of

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}=\mathbf{0}$$

$\boldsymbol{p}$ is outside the support polygon

$\boldsymbol{p}$ could NOT be the center of pressure, because all the GRFs are applied within the polygon, so that

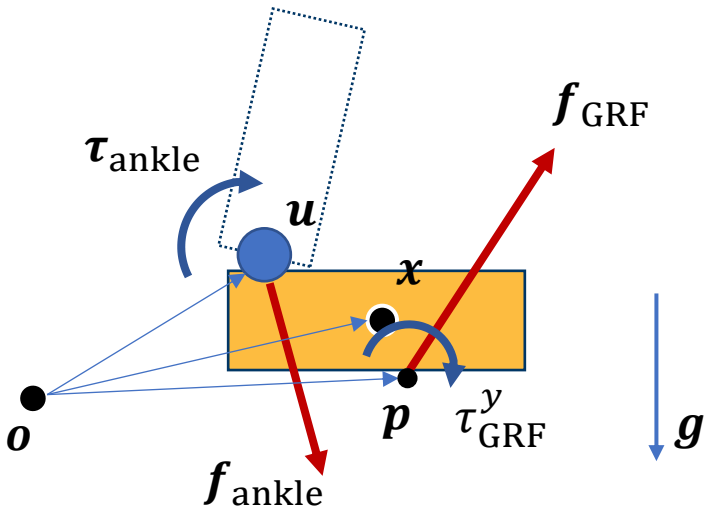
$$\tau^{xz}_{\text{GRF}}\neq\mathbf{0}$$

the foot will rotate…

Or, if $\boldsymbol{p}'$ is the real center of pressure, we have

$$\left((\boldsymbol{u}-\boldsymbol{o})\times\boldsymbol{f}_{\text{ankle}}\right)^{xz}+\left((\boldsymbol{p}'-\mathbf{o})\times\boldsymbol{f}_{\text{GRF}}\right)^{xz}+(\boldsymbol{x}-\boldsymbol{o})\times m\boldsymbol{g}+\boldsymbol{\tau}^{xz}_{\text{ankle}}\neq\mathbf{0}$$

Zero-Moment Point (ZMP)



The foot should not move in a **stance phase**

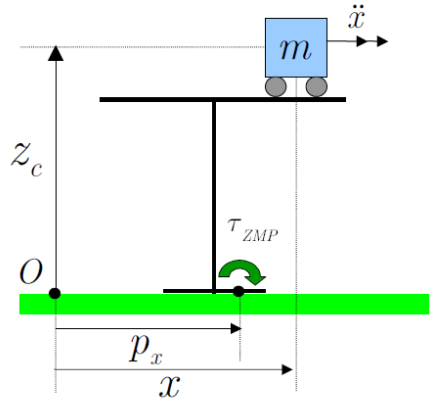
The existence of ZMP is an indication of dynamic balance

We can achieve balanced walking by controlling ZMP

But how?

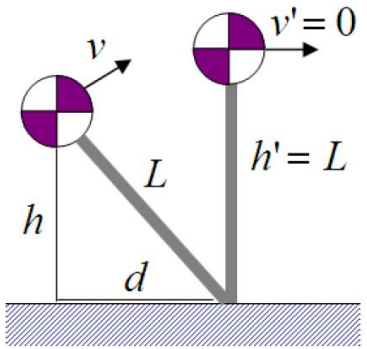
Simplified Models

- Simplify humanoid / biped robot into an abstract model
  - Often consists of a CoM and a massless mechanism
  - Need to map the state of the robot to the abstract model



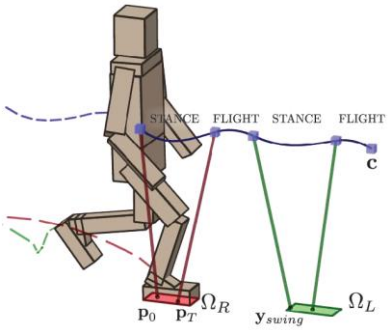
[Kajita et al. 2003]

- Plan the control and movement of the model
  - Optimization
  - Dynamic programming
  - Optimal control
  - MPC



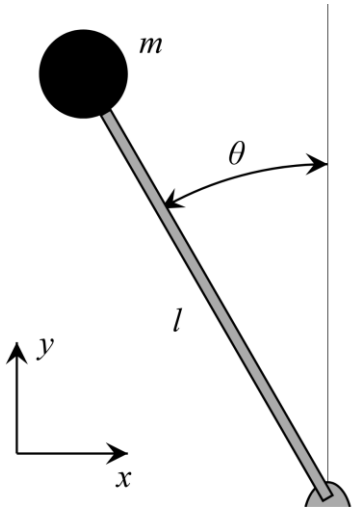
[Coros et al. 2010]

- Track the planned motion of the abstract model
  - Inverse Kinematics
  - Inverse Dynamics



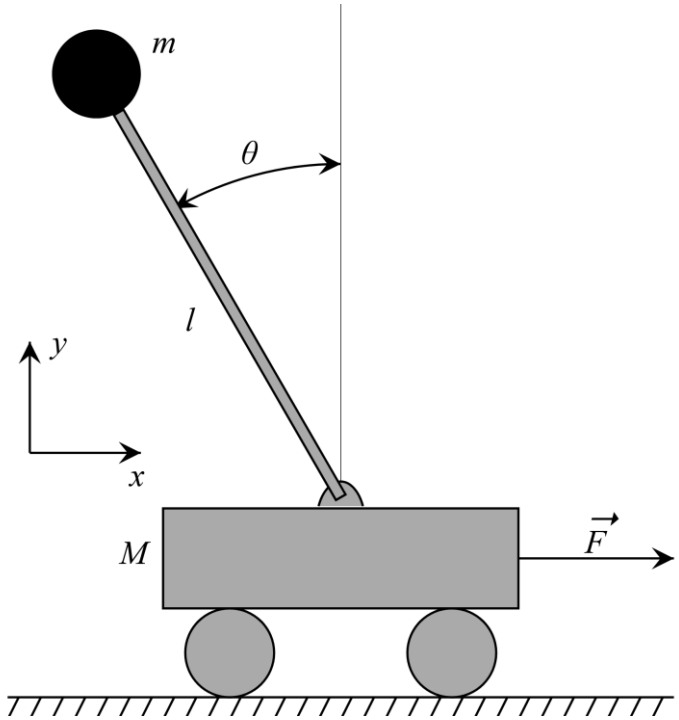
[Mordatch et al. 2010]

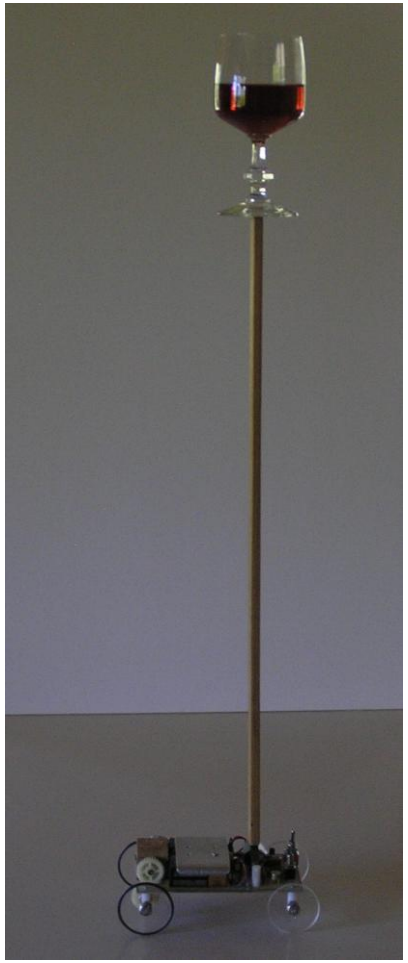
Inverted Pendulum Model (IPM)



$$\ddot{\theta} = \frac{g}{l}\sin\theta$$

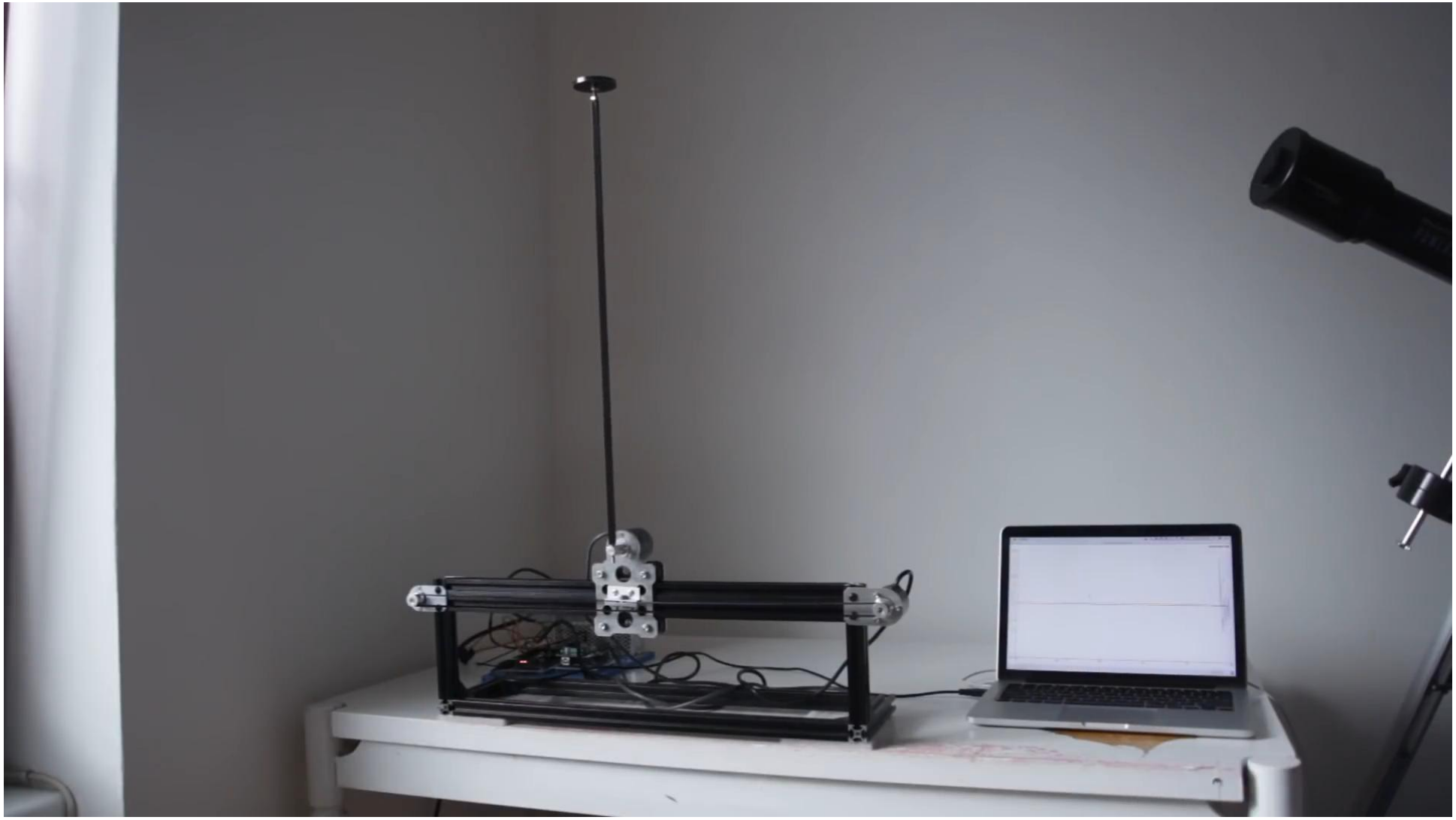
Inverted Pendulum Model (IPM)





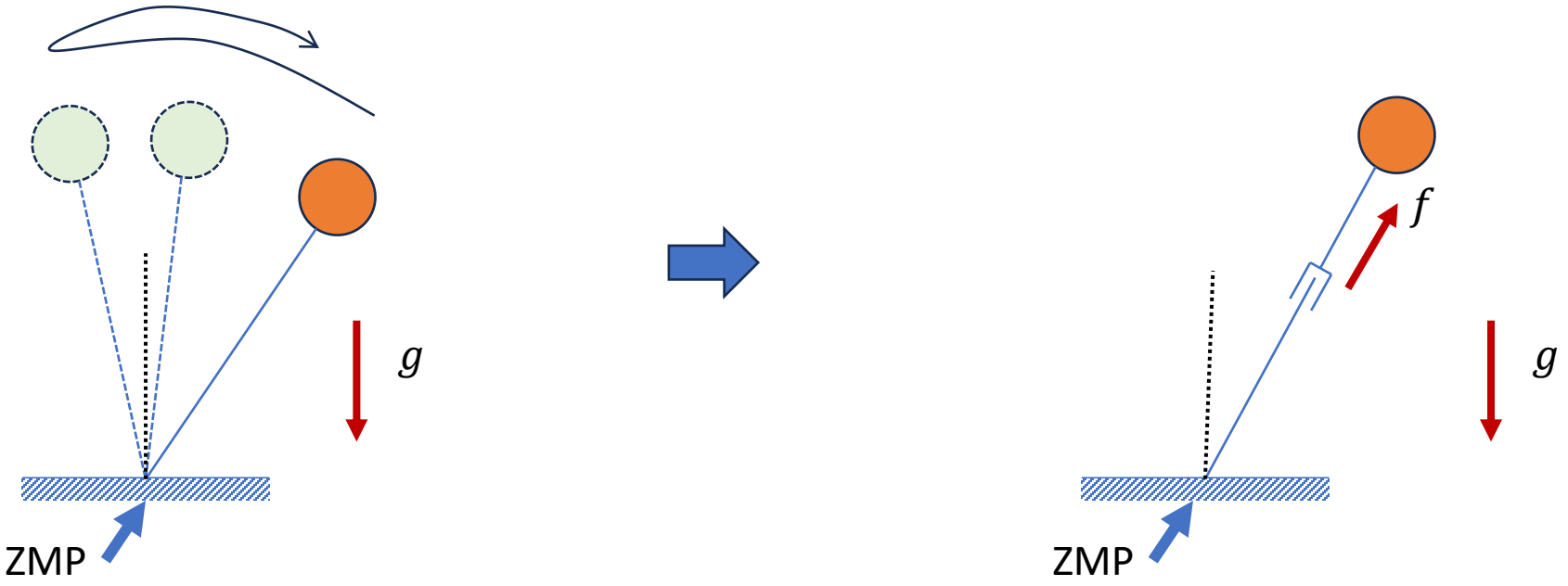
Inverted pendulum on a cart

Inverted Pendulum Model (IPM)

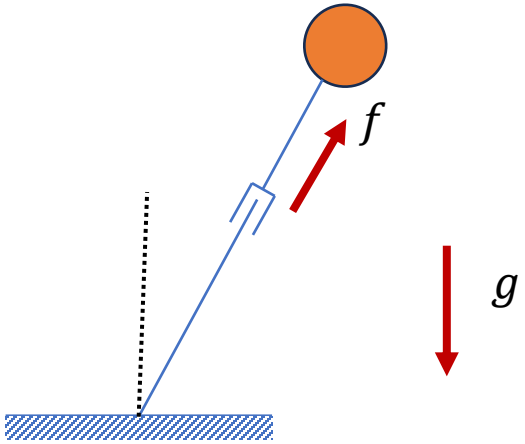


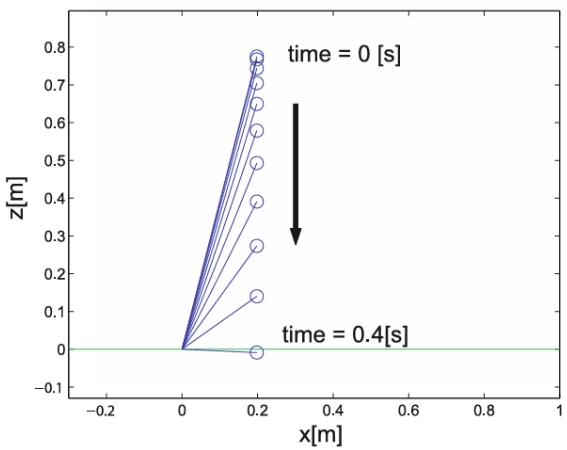
Inverted pendulum on a cart https://www.youtube.com/watch?v=nOSTzpA0nGk

Inverted Pendulum Model (LIPM)

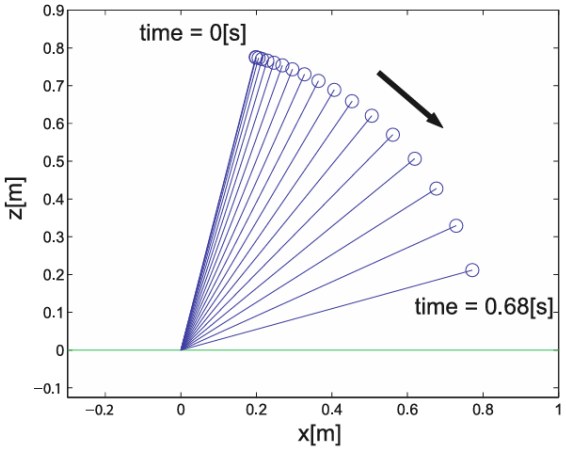


Inverted Pendulum Model (LIPM)

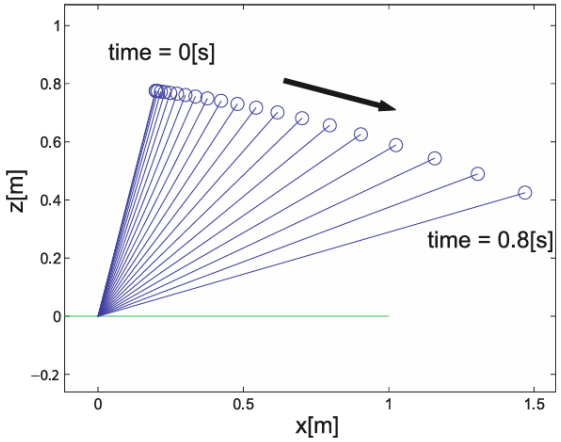




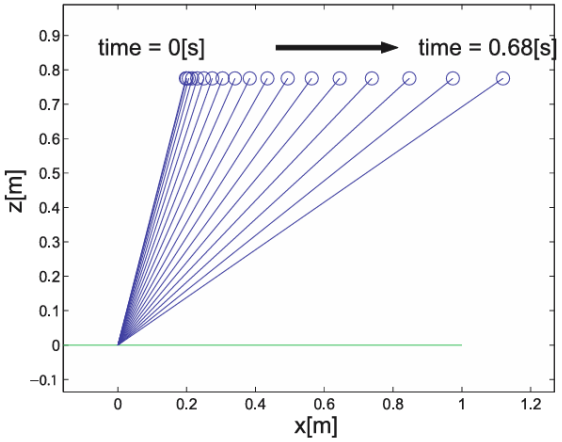
(a) $f = 0$: Free fall of CoM



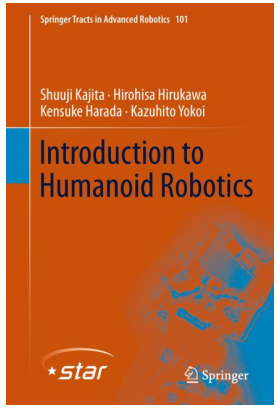
(b) $f = Mg\cos\theta - Mr\dot{\theta}^2$: Fall down with constant leg length



(c) $f = Mg$: Fall down and acceleration

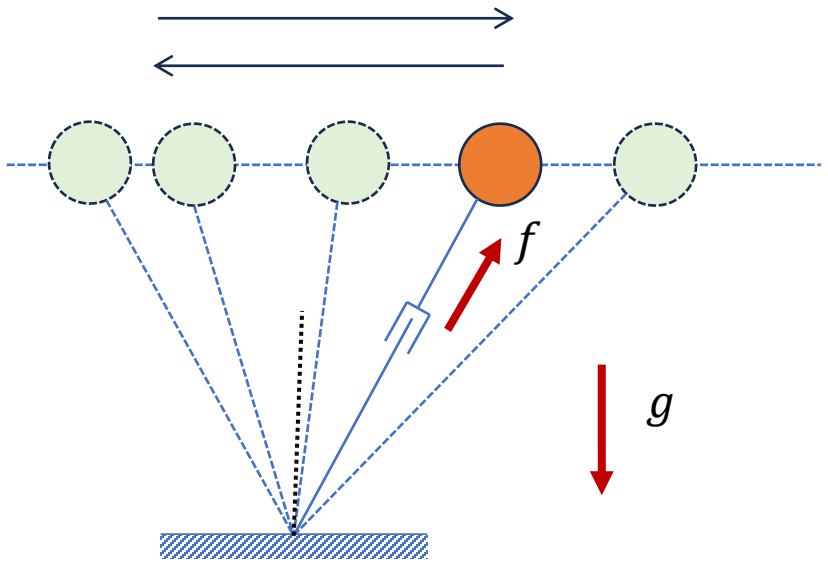


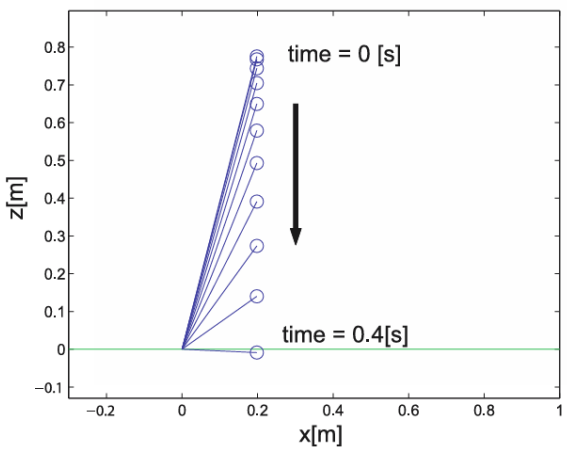
(d) $f = Mg/\cos\theta$: CoM accelerates while keeping the initial height



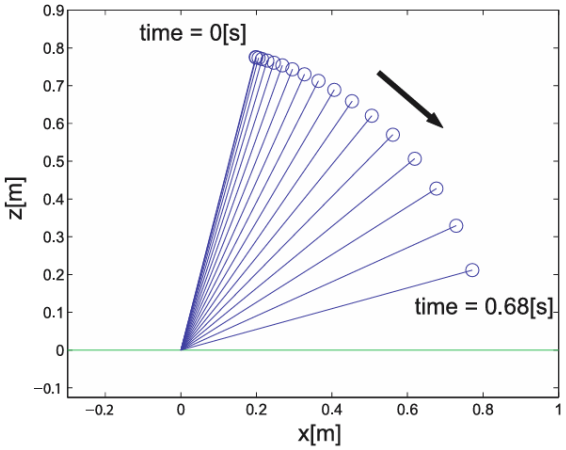
[Kajita et al. Introduction to Humanoid Robotics]

Inverted Pendulum Model (LIPM)

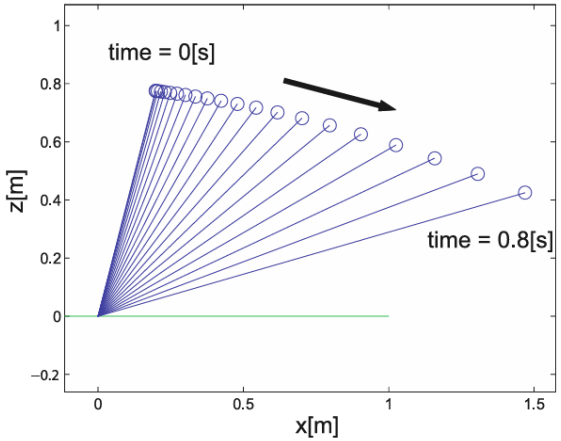




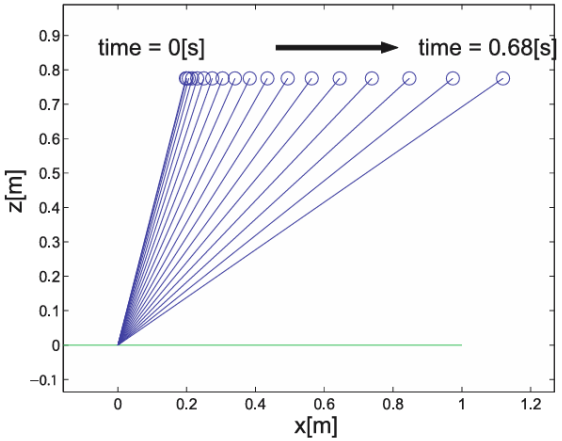
(a) $f = 0$: Free fall of CoM



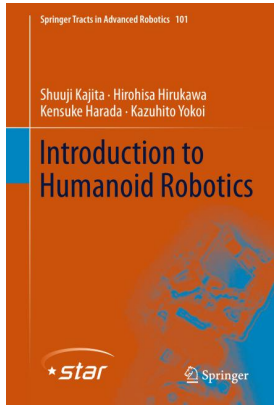
(b) $f = Mg\cos\theta - Mr\dot{\theta}^2$: Fall down with constant leg length



(c) $f = Mg$: Fall down and acceleration

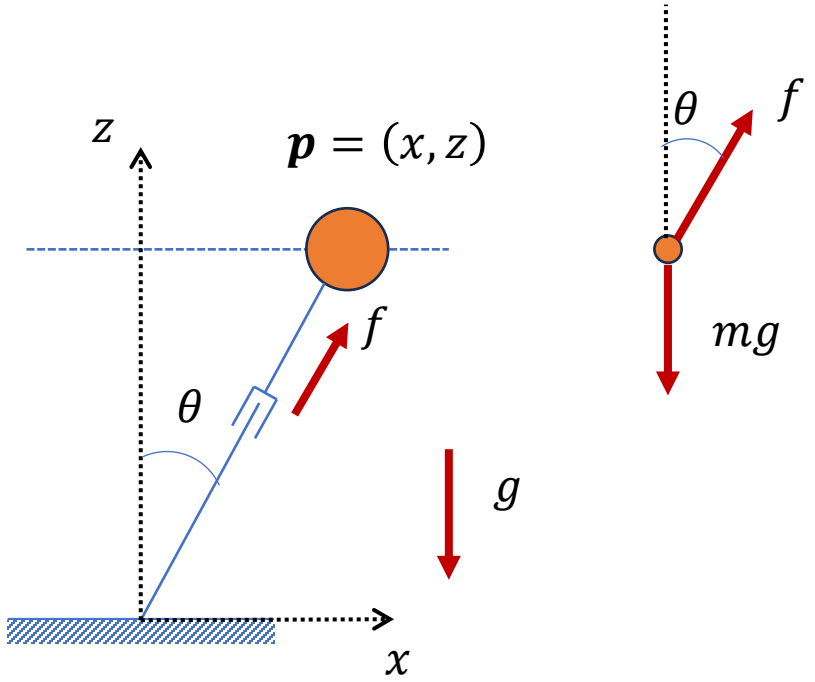


(d) $f = Mg/\cos\theta$: CoM accelerates while keeping the initial height



[Kajita et al. Introduction to Humanoid Robotics]

Inverted Pendulum Model (LIPM)



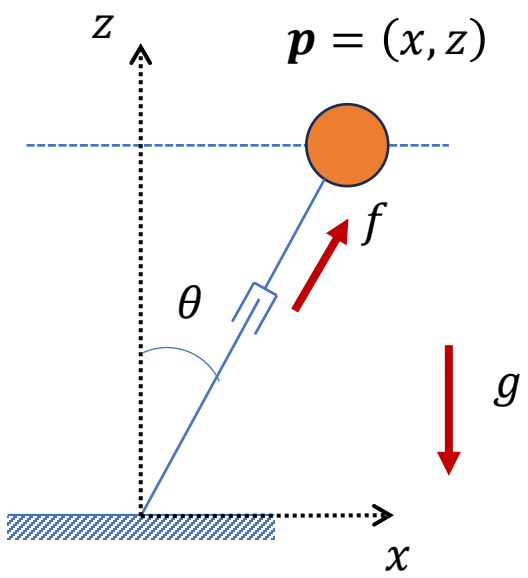
$$m\ddot{x} = f \sin\theta$$

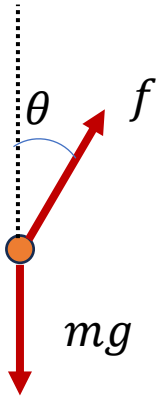
$$f \cos\theta = mg$$

$$m\ddot{x} = mg \tan\theta = mg\frac{x}{z}$$

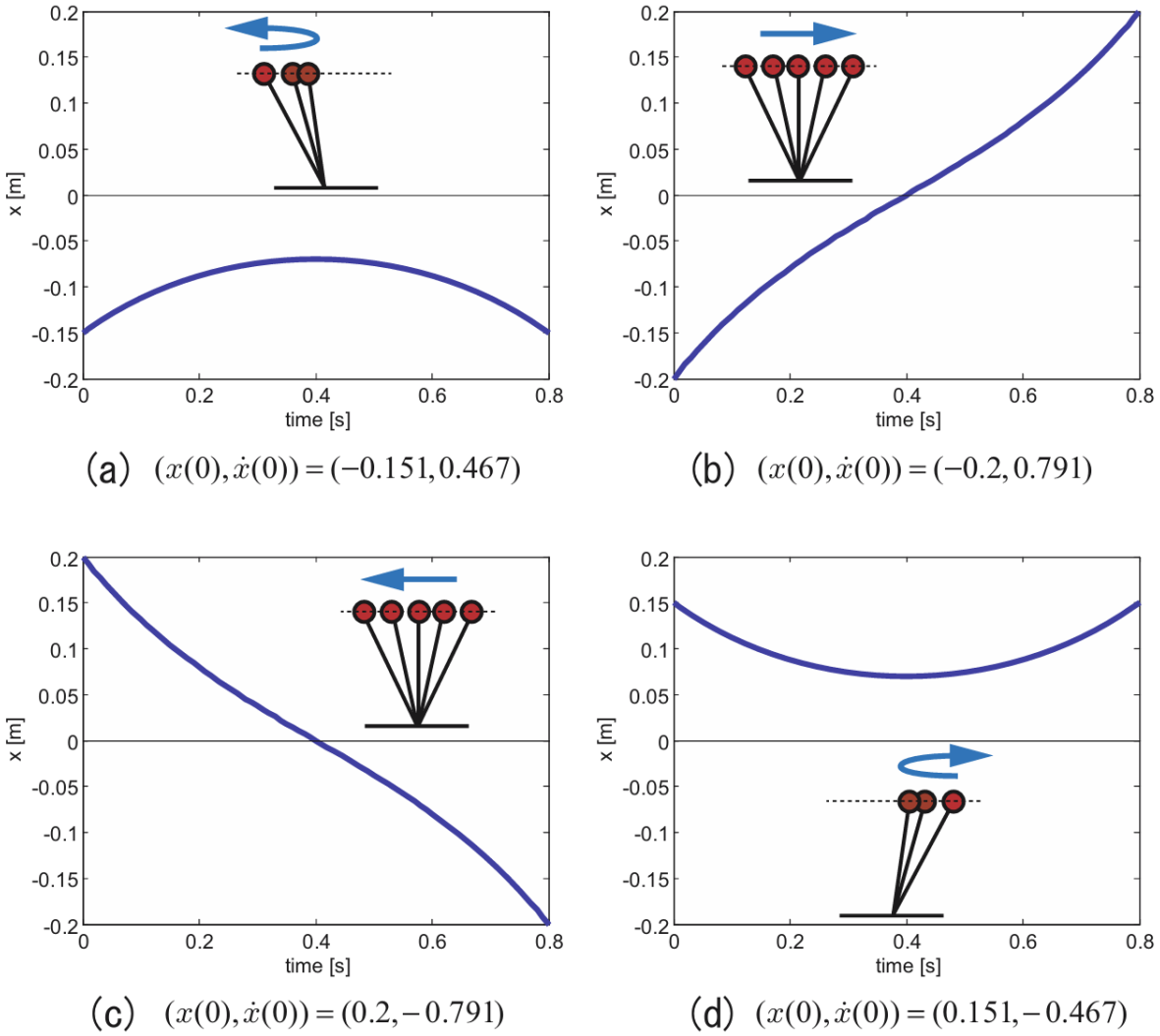
$$\ddot{x} = \frac{g}{z}x$$

Inverted Pendulum Model (LIPM)

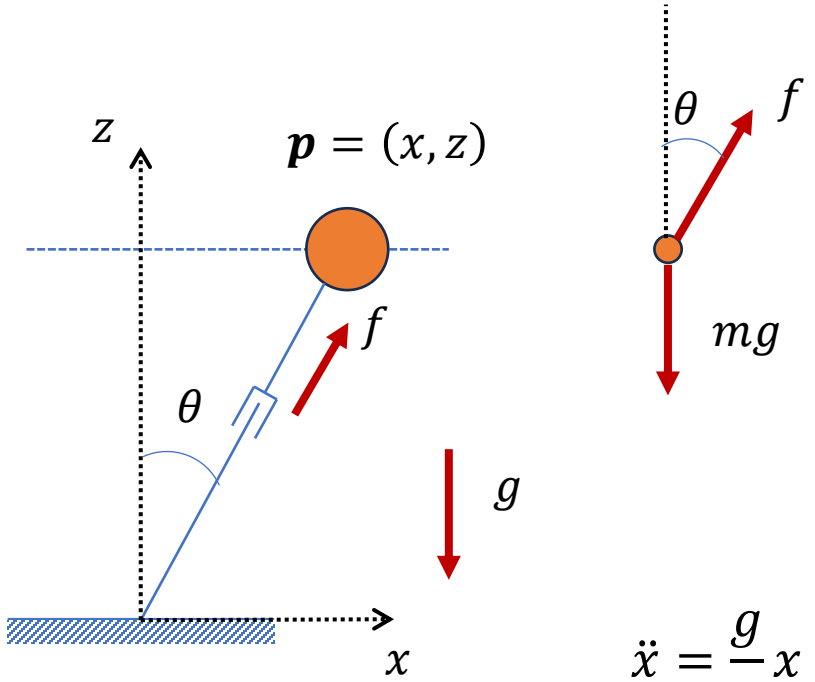




$$\ddot{x} = \frac{g}{z} x$$



Inverted Pendulum Model (LIPM)

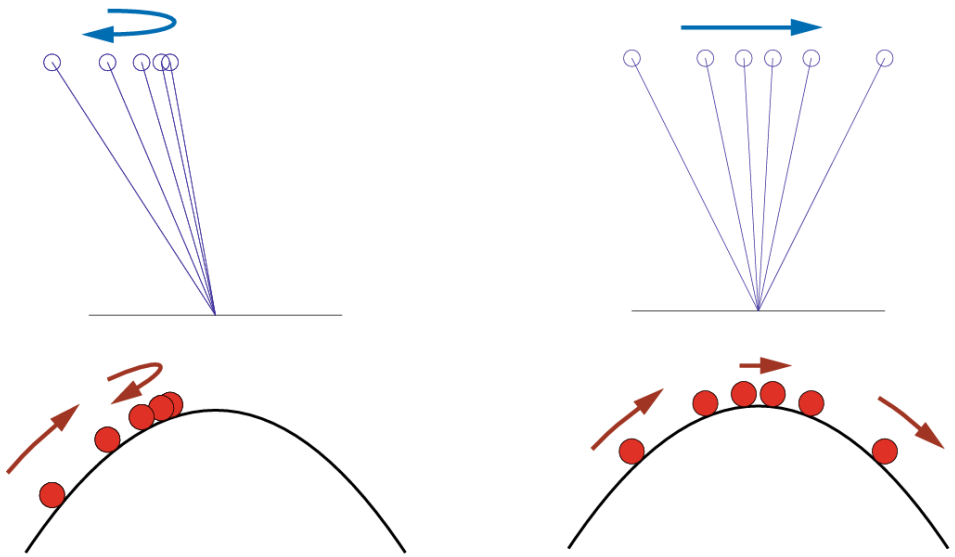


$$\ddot{x} = \frac{g}{z} x$$

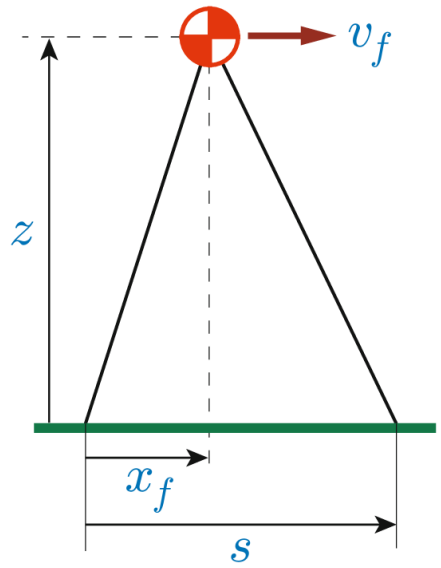
Kinetic energy

orbital energy $\quad E = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\frac{g}{z}x^2$

imaginary "potential" energy



Simple Biped Gait with LIPM



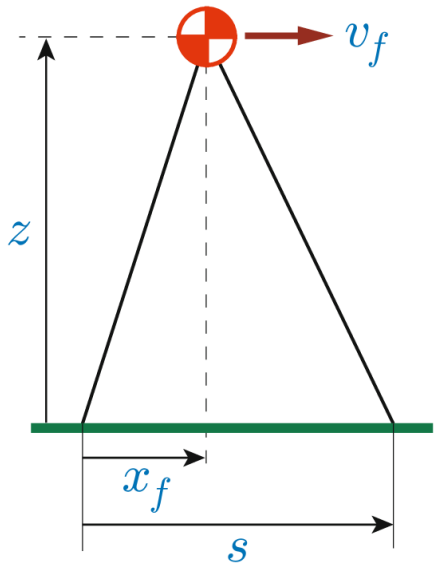
Before switch:

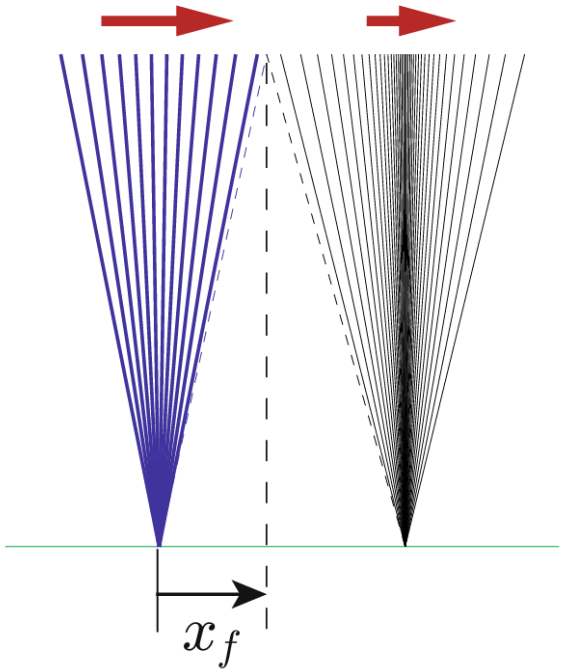
$$E_1 = \frac{1}{2} v_f^2 - \frac{g}{2z} x_f^2$$

After switch:

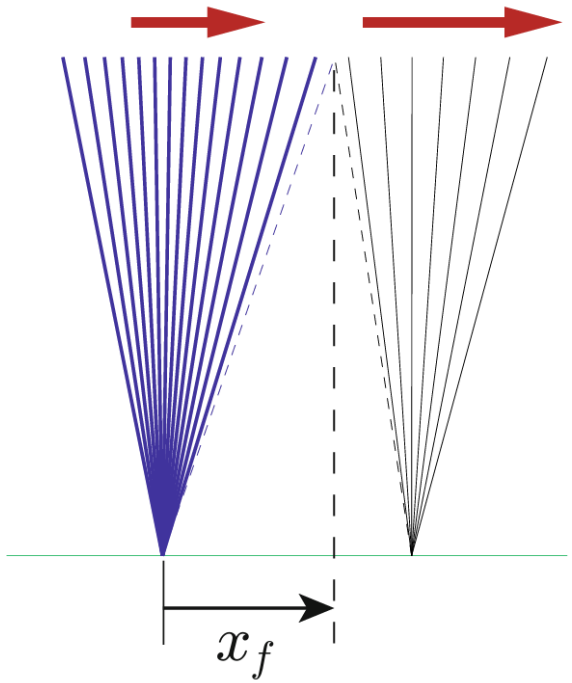
$$E_2 = \frac{1}{2} v_f^2 - \frac{g}{2z} \left(x_f - s\right)^2$$

Simple Biped Gait with LIPM



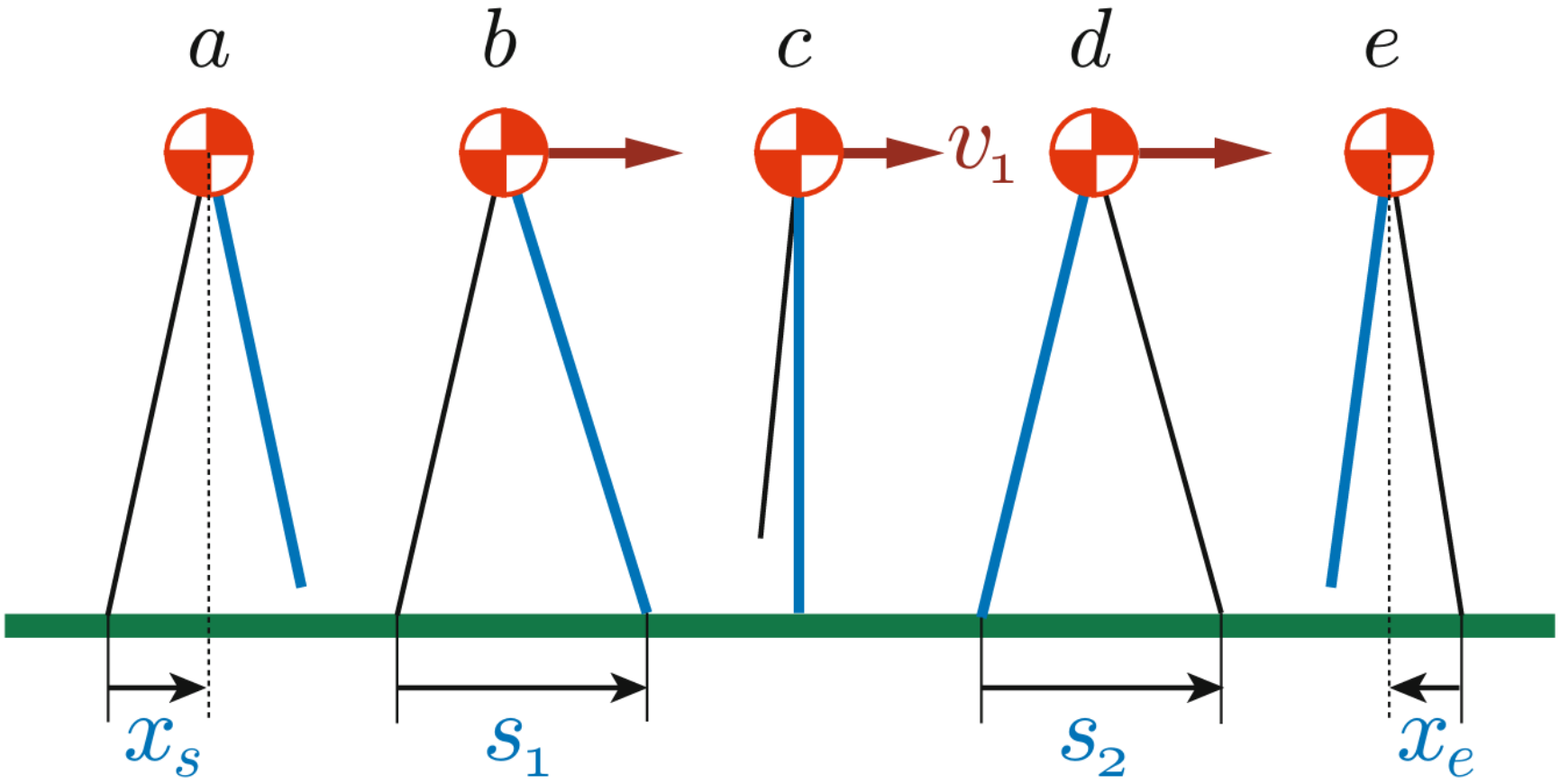


Early switch: slow down



Late switch: speed up

Simple Biped Gait with LIPM



Walking == Falling + Step Planning

Passive Dynamic Walker

Inverted Pendulum Model (IPM)

- Step Plan with IPM

Generalized Biped Walking Control

Stelian Coros Philippe Beaudoin Michiel van de Panne*

University of British Columbia

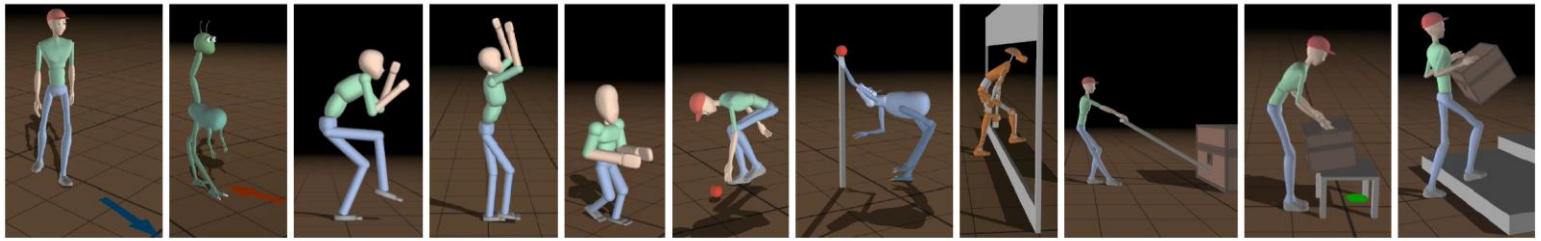
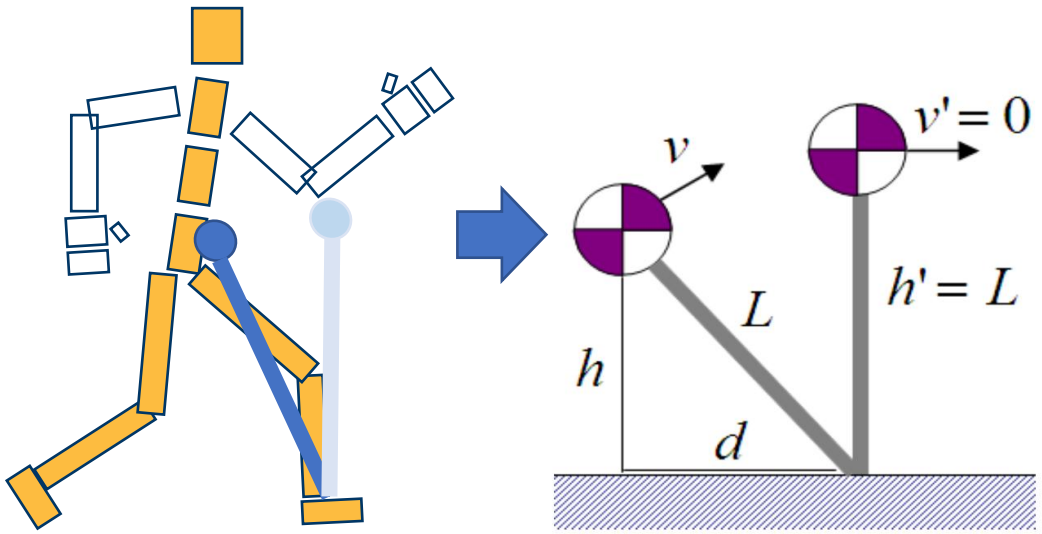


Figure 1: *Real-time physics-based simulation of walking. The method provides robust control across a range of gaits, styles, characters, and skills. Motions are easily authored by novice users.*

[Coros et al. 2010 - **Generalized Biped Walking Control**]

Inverted Pendulum Model (IPM)

- Step Plan with IPM
  - Map CoM of the character and the stance foot as IPM
  - Plan the position of the next foot step so that the mass point rests at the top of the pendulum
  - Create foot trajectory based on the step plan
  - Compute target poses using IK



[Coros et al. 2010 - **Generalized Biped Walking Control**]

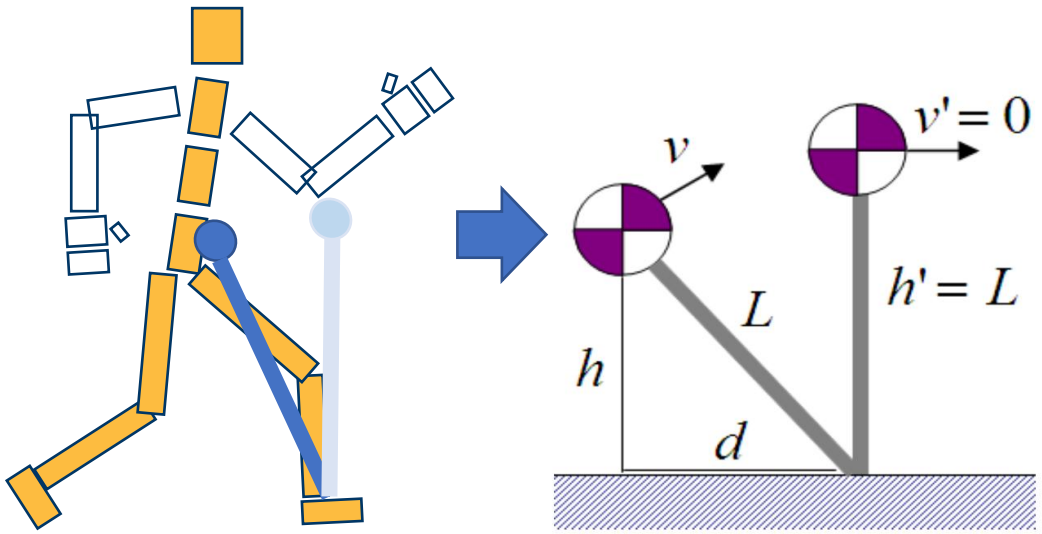
Inverted Pendulum Model (IPM)

- Step Plan with IPM

$$\frac{1}{2}mv^2 + mgh = \frac{1}{2}mv'^2 + mgh'$$

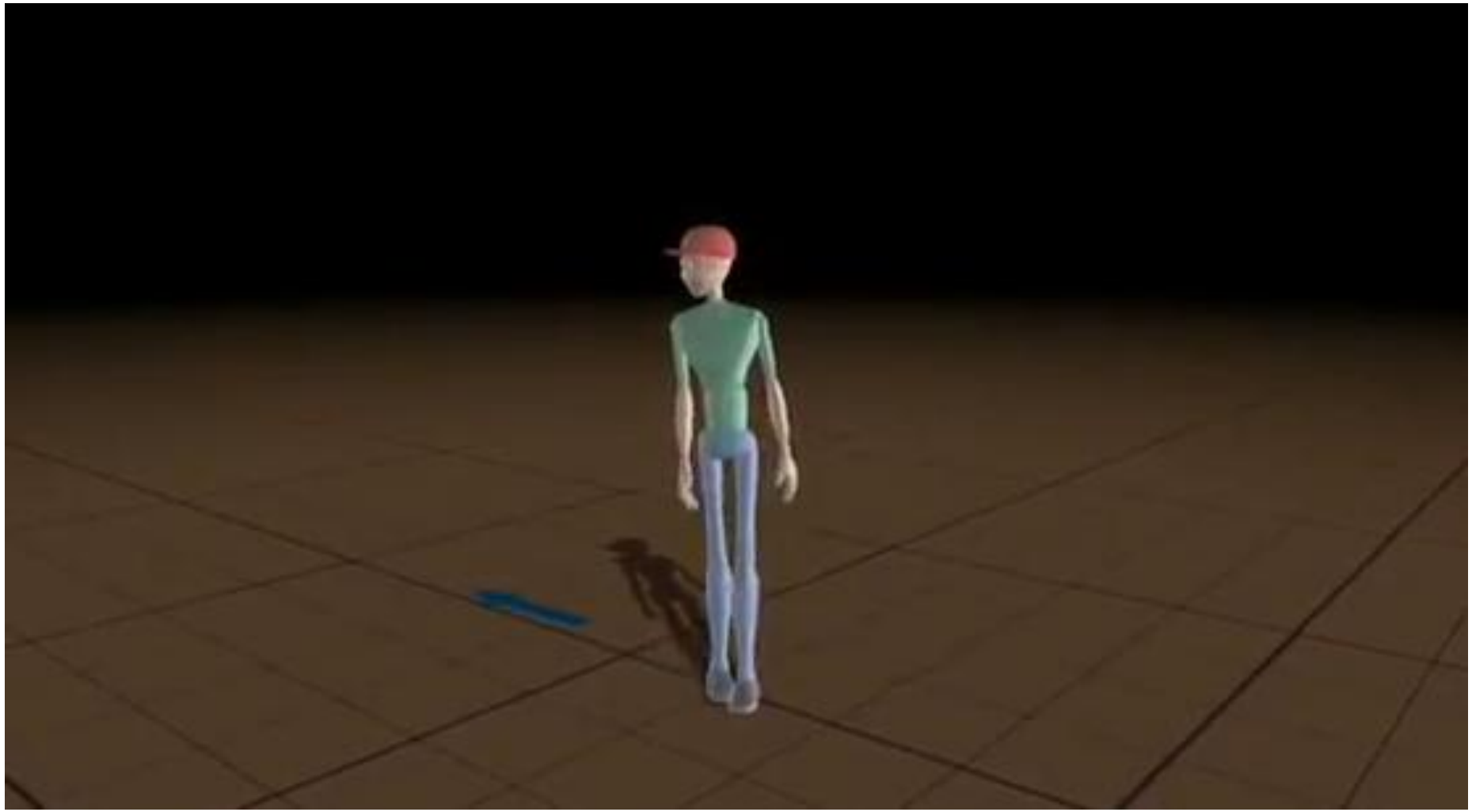
$$v' = 0 \text{ and } h' = L = \sqrt{h^2 + d^2}$$

$$d = v\sqrt{h/g + v^2/(4g^2)}.$$



[Coros et al. 2010 - **Generalized Biped Walking Control**]

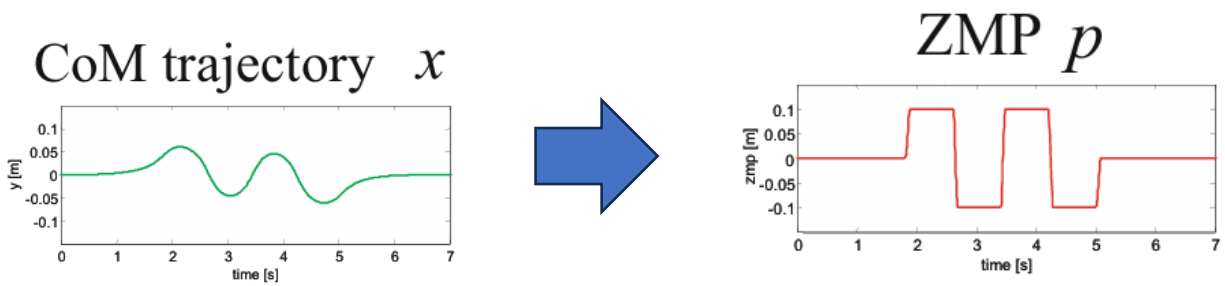
Generalized walking control

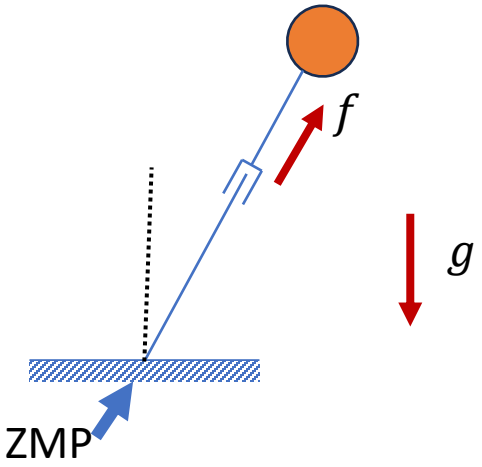


[Coros et al. 2010]

Problem of IPM-based Gait Planning

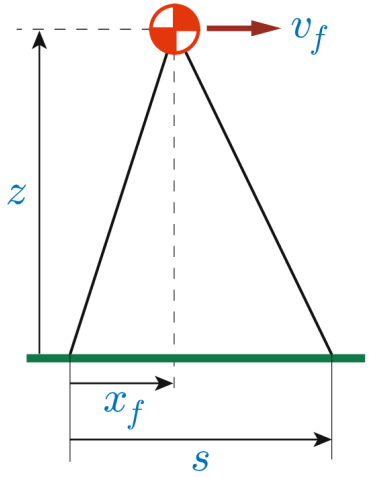
- Given a desired CoM trajectory
- Compute the ZMP trajectory





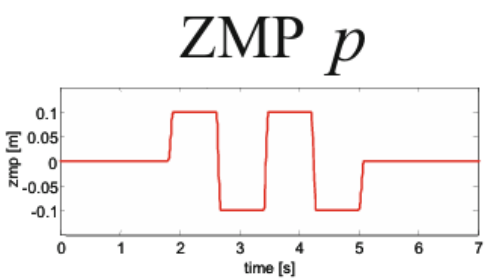
- ZMP trajectory <=> Footstep plan

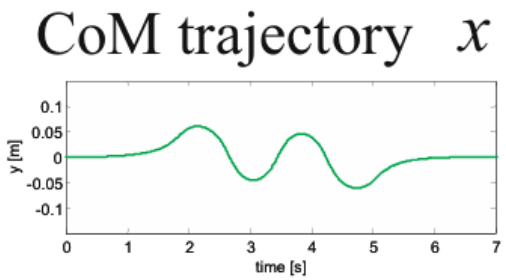
- What if we need specific footsteps?

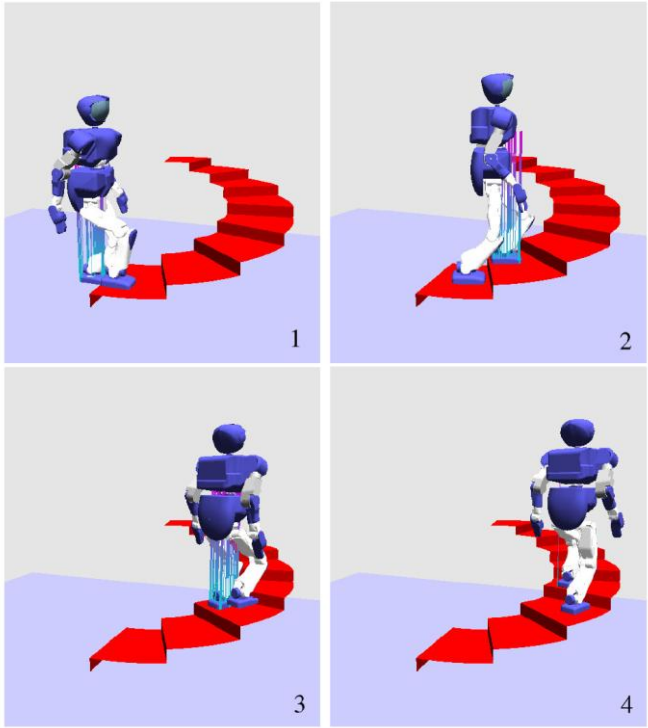


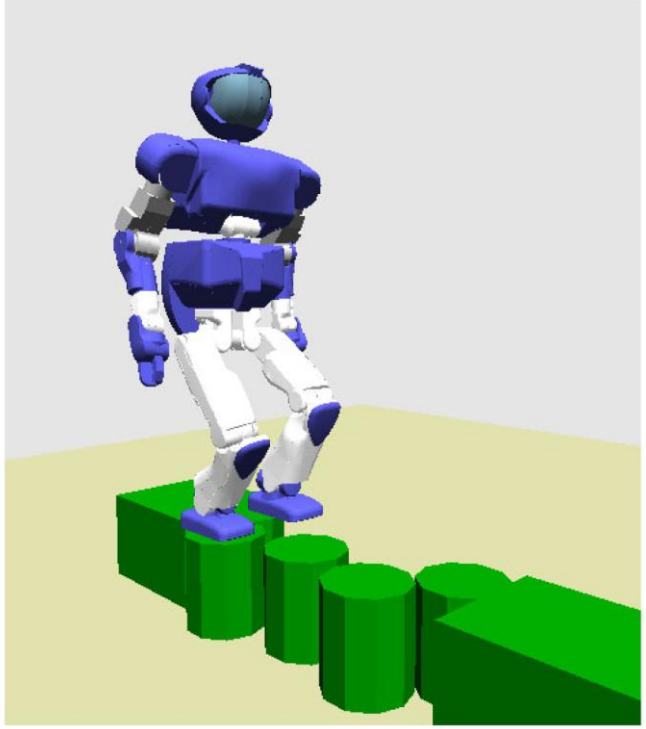
ZMP-Based Walking Pattern Generation

- Given a desired ZMP trajectory
- Compute the CoM trajectory



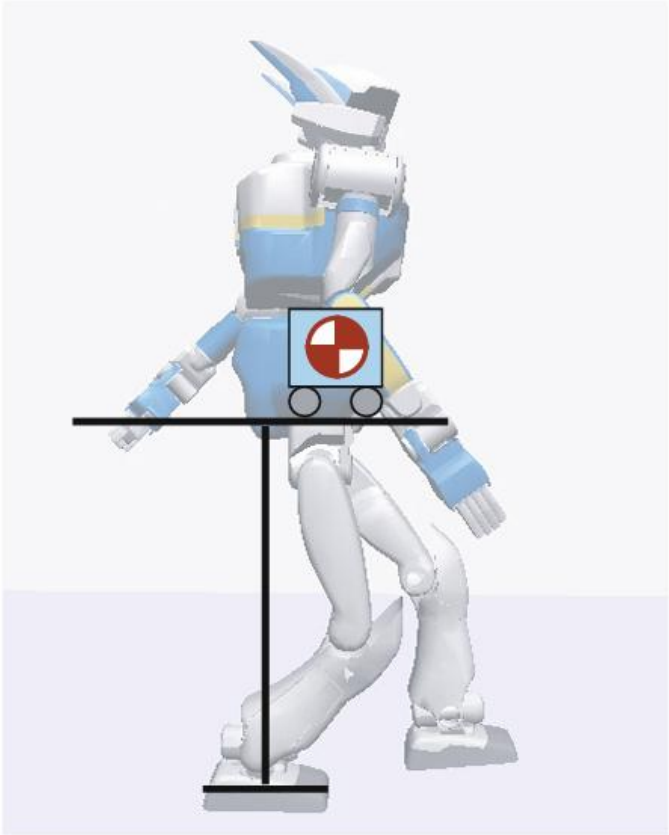


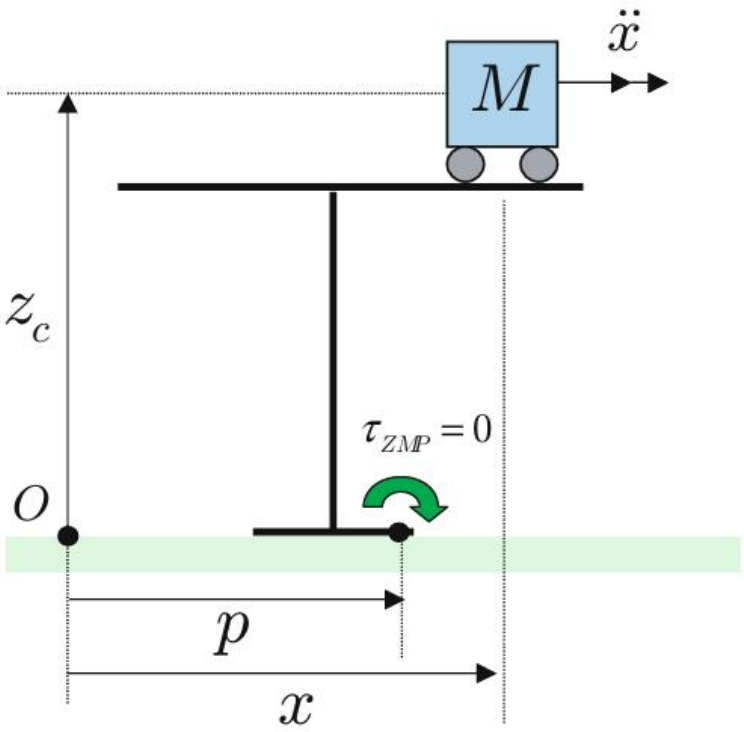




[Kajita et al. 2003]

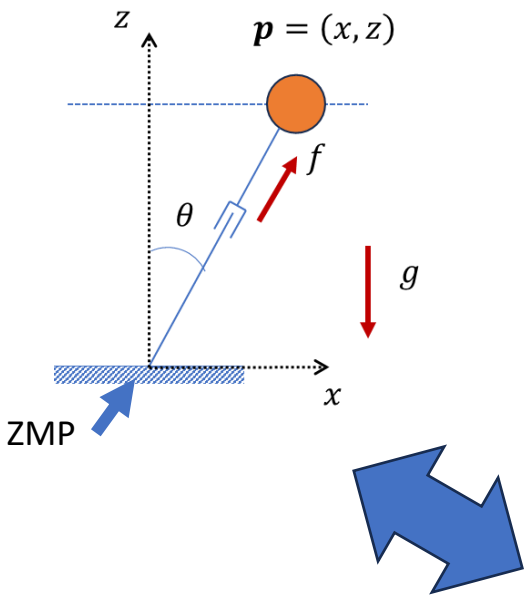
Cart-Table Model

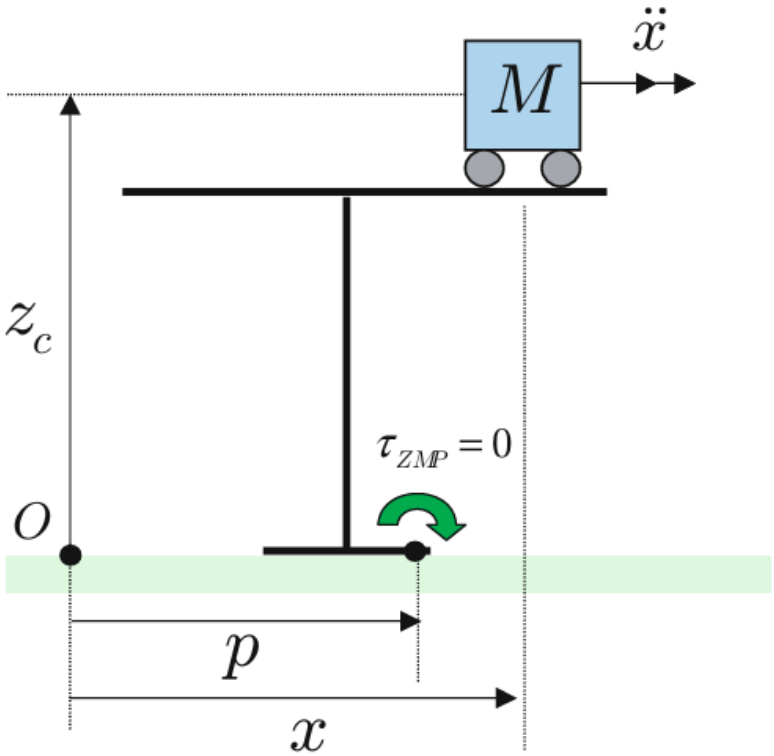




A mass-less table with a cart moving on it

Cart-Table Model





The LIPM:

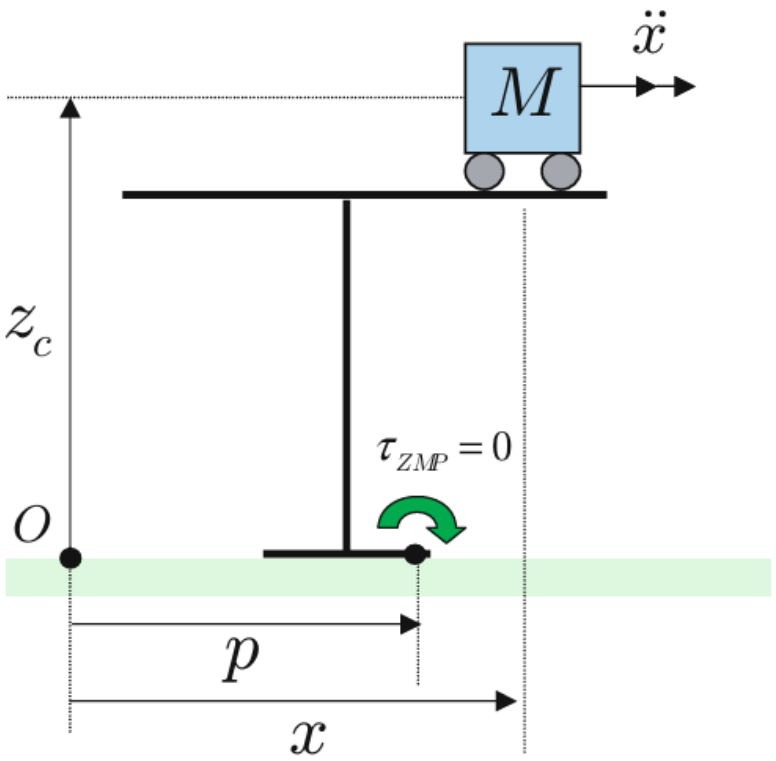
$$\ddot{x} = \frac{g}{z}(x - p)$$

The ZMP:

$$p_{zmp} = x - \frac{z_c}{g}\ddot{x}$$

The table will turn over unless the cart moves with a proper $\ddot{x}$

Cart-Table Model



The ZMP:

$$p_{zmp} = x - \frac{z_c}{g}\ddot{x}$$

Using approximation

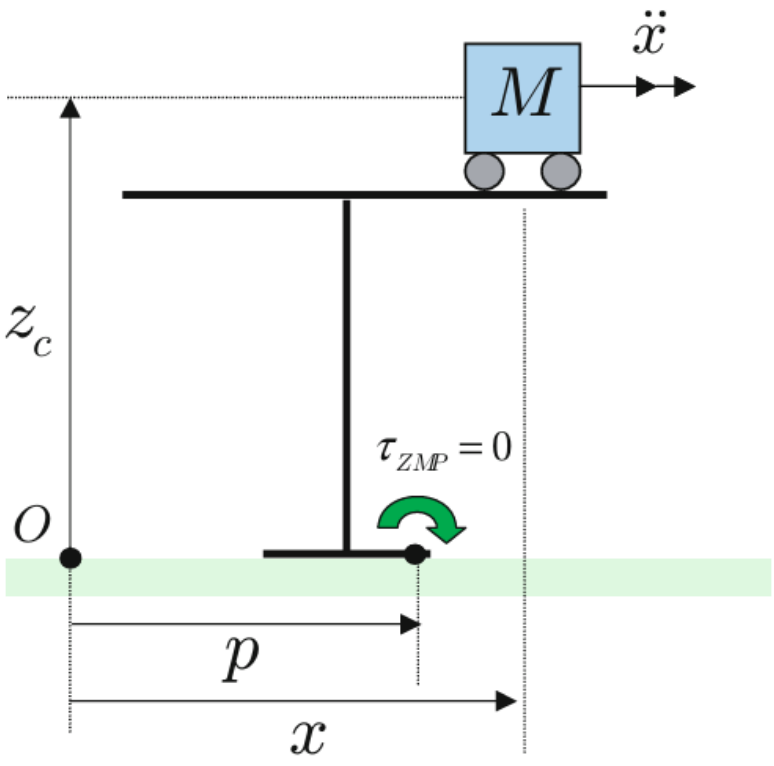
$$\ddot{x} = \frac{x_{i-1} - 2x_i + x_{i+1}}{\Delta t^2}$$

We have:

$$p_i = x_i - \frac{z_c}{g}\frac{x_{i-1} - 2x_i + x_{i+1}}{\Delta t^2}$$

$$= a_i x_{i-1} + b_i x_i + c_i$$

Cart-Table Model



Discrete ZMP equation:

$$p_i = a_i x_{i-1} + b_i x_i + c_i$$

Given ZMP trajectory

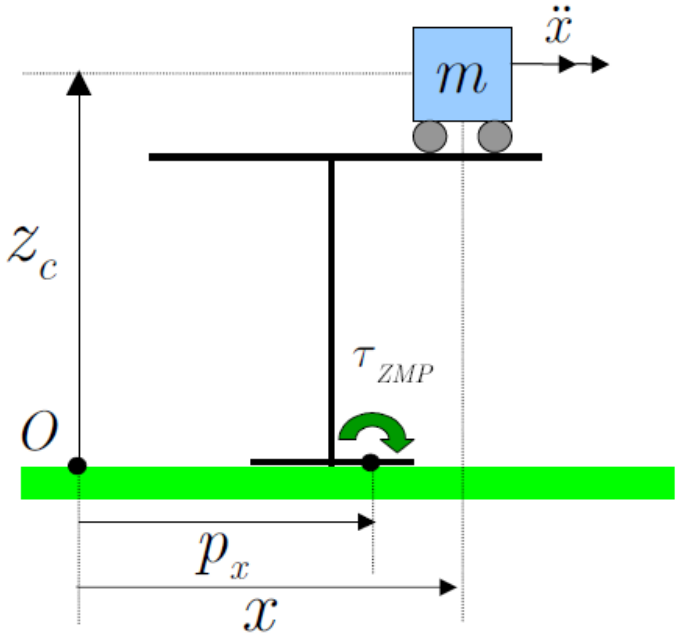
$$\{p_1, p_2, \dots p_{N-1}\}$$

We can solve for CoM Trajectory

$$\{x_0, x_2, \dots x_N\}$$

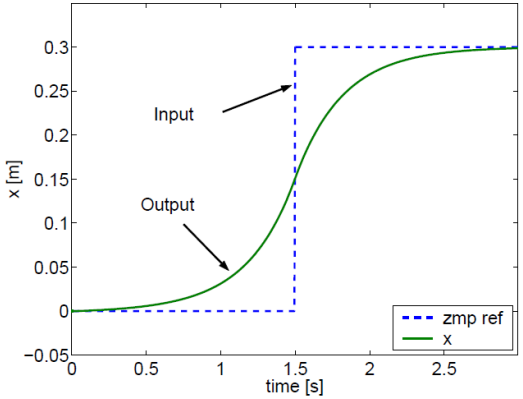
$$\begin{bmatrix} \vdots \\ p_1 \\ \vdots \\ p_{N-1} \\ \vdots \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 & & & \\ & a_2 & b_2 & c_2 & \dots & \\ & & & \dots & \dots & \\ & & & a_{N-1} & b_{N-1} & c_{N-1} \end{bmatrix} \begin{bmatrix} x_0 \\ \vdots \\ x_N \end{bmatrix}$$

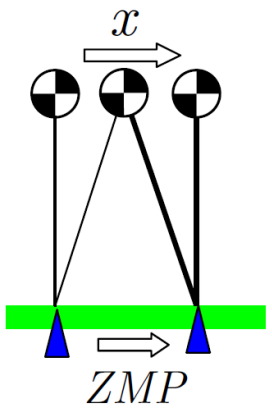
ZMP-Based Walking Pattern Generation

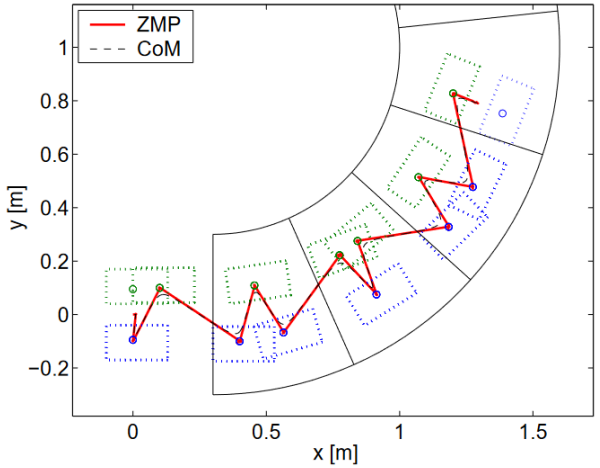


Biped Walking Pattern Generation by using Preview Control of Zero-Moment Point

Shuuji KAJITA, Fumio KANEHIRO, Kenji KANEKO, Kiyoshi FUJIWARA, Kensuke HARADA, Kazuhito YOKOI and Hirohisa HIRUKAWA

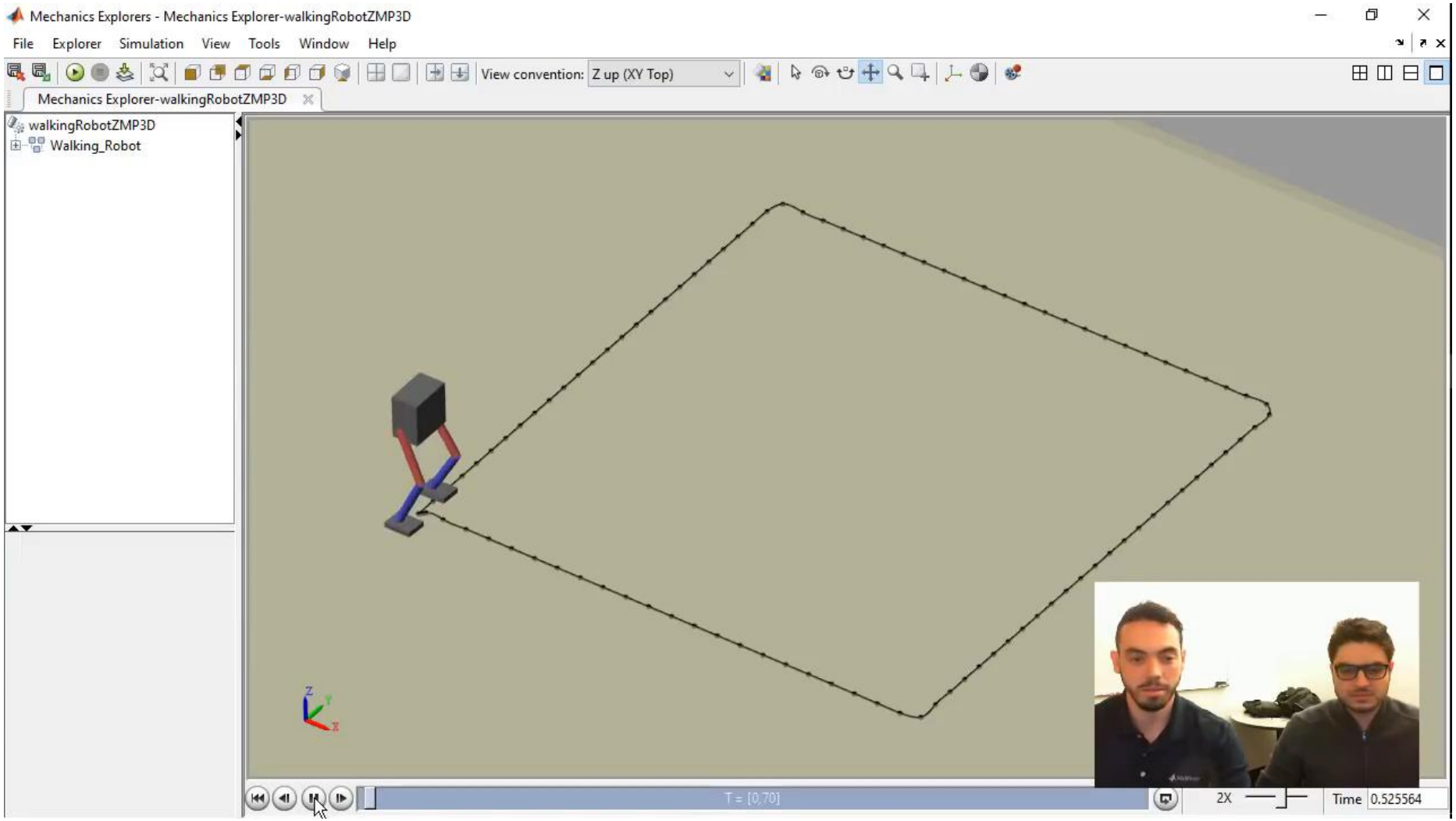






[Kajita et al. 2003]

ZMP-Based Walking Pattern Generation



An Interesting talk: https://www.youtube.com/watch?v=axK-YVEGReo

ZMP-Based Walking Pattern Generation

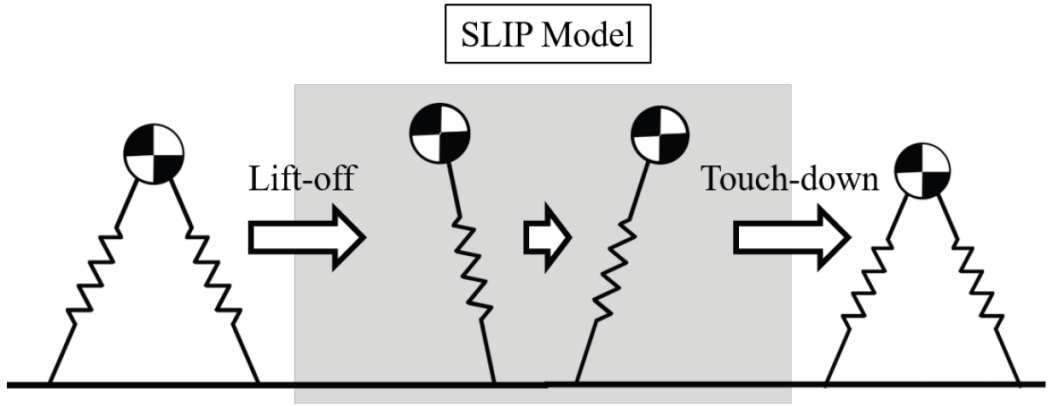
Biped Robot Walking using ZMP Preview Control(2022.09.08)

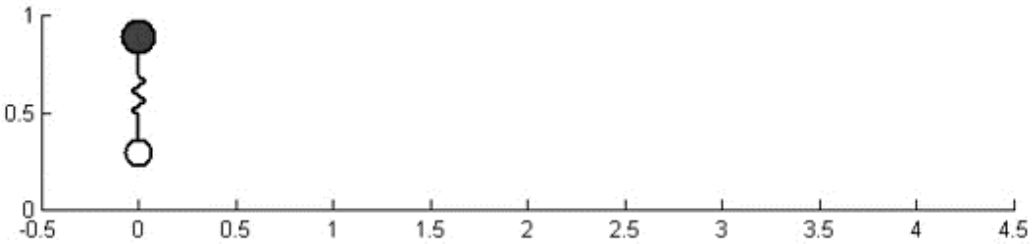
https://www.youtube.com/shorts/z1Tz3Hsx9II

ASIMO

Inverted Pendulum Model (IPM)

- Spring-Loaded Inverted Pendulum (SLIP)





SLIP running explained
https://www.youtube.com/watch?v=ZccJ_V2WR8k

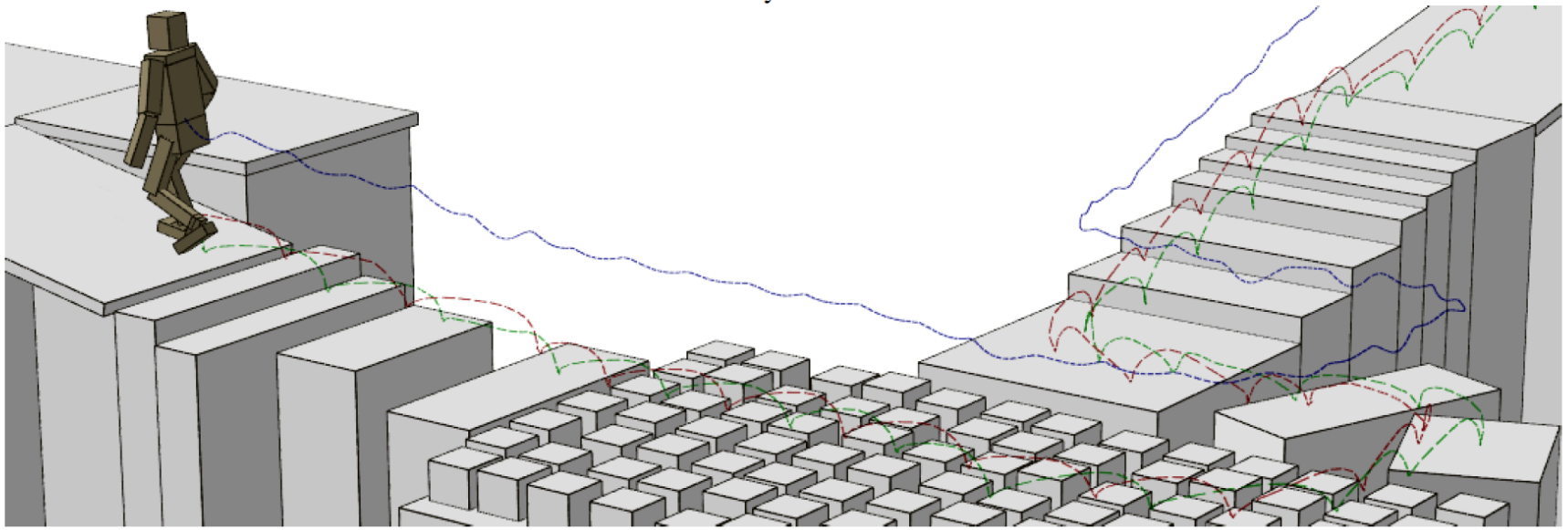
Inverted Pendulum Model (IPM)

- Spring-Loaded Inverted Pendulum (SLIP)

Robust Physics-Based Locomotion Using Low-Dimensional Planning

Igor Mordatch Martin de Lasa Aaron Hertzmann

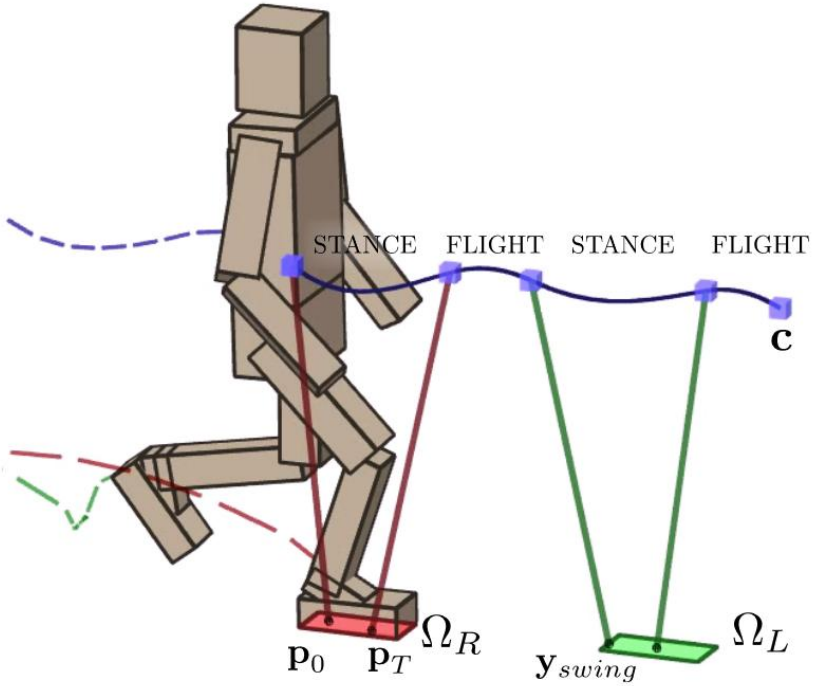
University of Toronto



[Mordatch et al. 2010 - **Robust Physics-Based Locomotion Using Low-Dimensional Planning**]

Inverted Pendulum Model (IPM)

- Spring-Loaded Inverted Pendulum (SLIP)



Given the optimal preview motion $\mathbf{S}^\star$, we calculate:

$$\mathbf{x} = \begin{bmatrix} \boldsymbol{\tau}^T & \ddot{\mathbf{q}}^T & \boldsymbol{\lambda}^T \end{bmatrix}^T$$

$$\mathbf{x}^* = \arg\min_{\mathbf{x}} \sum_i w_i E_i(\mathbf{x})$$

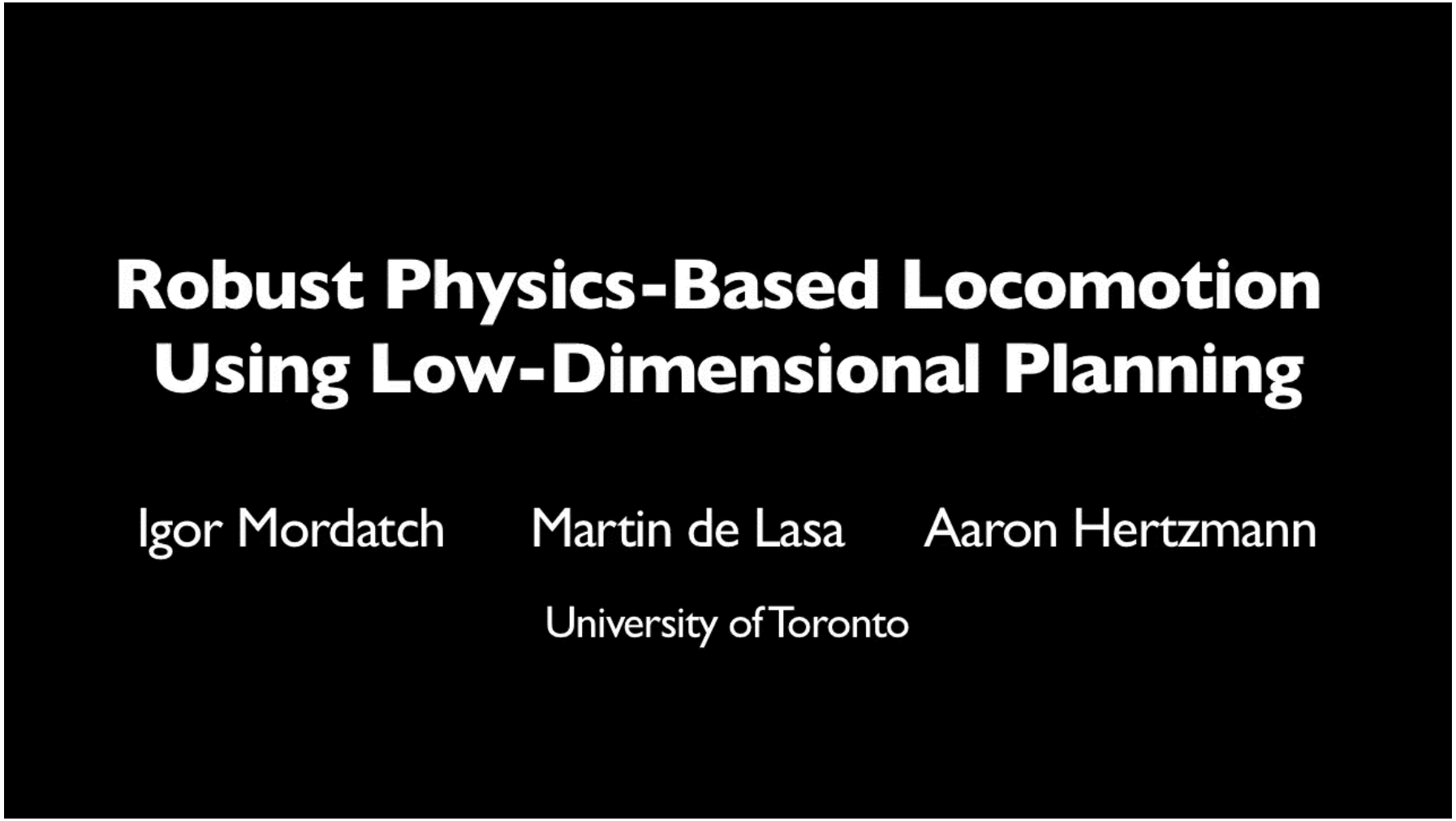
$$\text{subject to } C(\mathbf{x}) = 0, \mathbf{D}\mathbf{x} + \mathbf{f} \geq 0$$

$$E_{com} = ||\ddot{\mathbf{c}}_r - \ddot{\mathbf{c}}||^2$$

$$E_{heading} = (\ddot{\alpha}_r - \ddot{\alpha})^2$$

[Mordatch et al. 2010 - **Robust Physics-Based Locomotion Using Low-Dimensional Planning**]

Inverted Pendulum Model (IPM)



[Mordatch et al. 2010 - **Robust Physics-Based Locomotion Using Low-Dimensional Planning**]

SIMBICON

- SIMBICON (SIMple BIped Locomotion CONtrol)
  - Yin et al. 2007

SIMBICON: Simple Biped Locomotion Control

KangKang Yin Kevin Loken Michiel van de Panne*

University of British Columbia

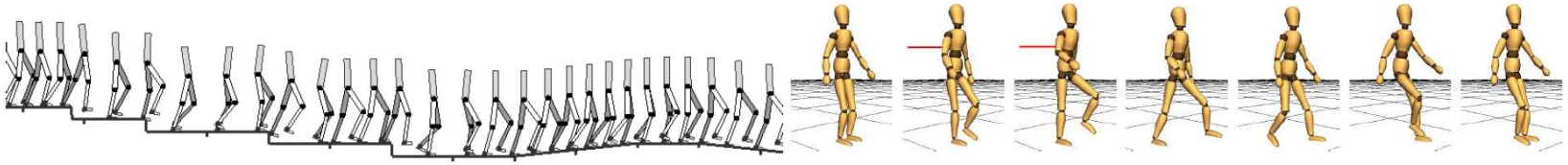
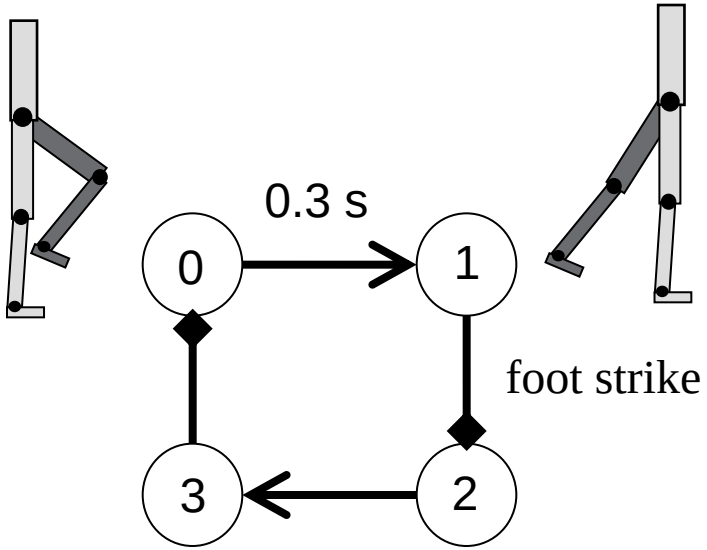
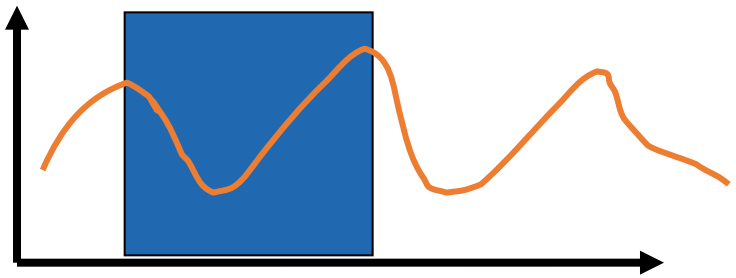


Figure 1: *Real-time physics-based character simulation with our framework. (a) A single controller for a planar biped responds to unanticipated changes in terrain. (b) A walk controller reconstructed from motion capture data responds to a* $350N, 0.2s$ *diagonal push to the torso.*

SIMBICON

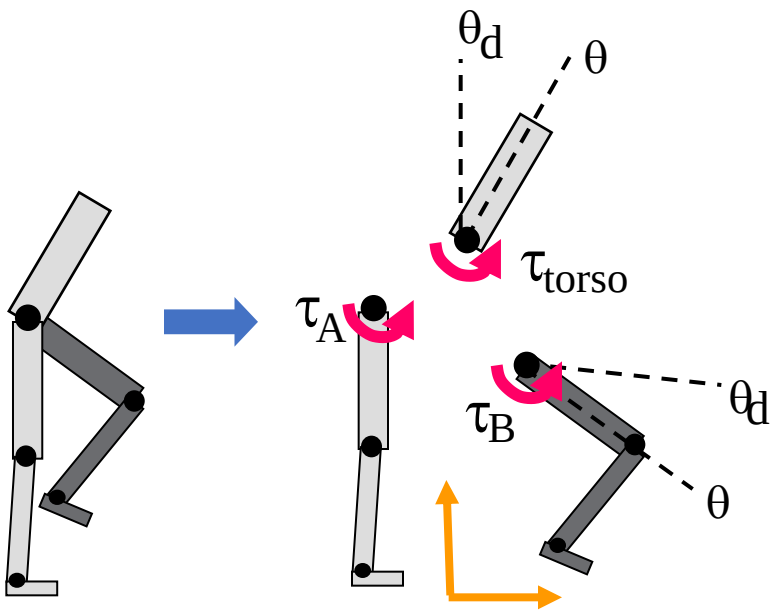
- Step 1: develop a cyclical base motion
  - PD controllers track target angles
  - FSM (Finite State Machine) or mocap





SIMBICON

- Step 2:
  - control torso and swing-hip wrt world frame

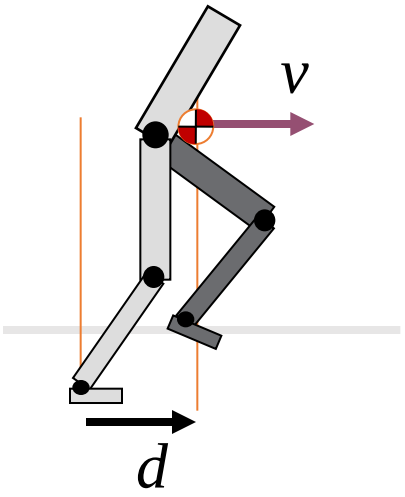


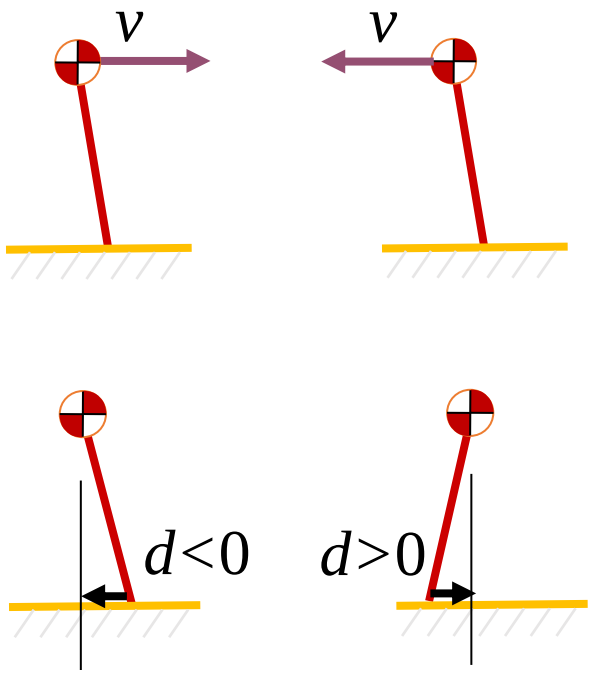
$$\tau_A = -\tau_{\text{torso}} - \tau_B$$

Newton's third law

SIMBICON

- Step 3: COM feedback



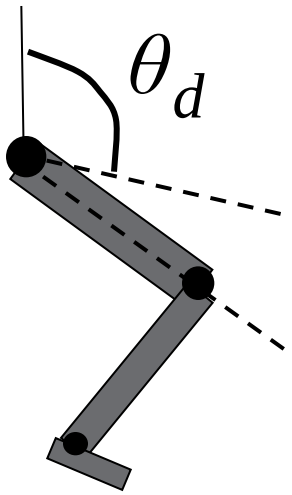


COM velocity matters

COM position matters

SIMBICON

- Step 3: COM feedback



Swing Leg

base controller

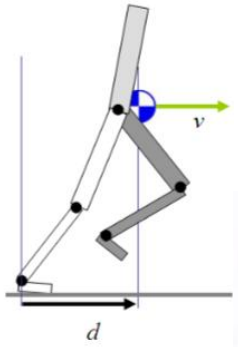
continuous feedback

$$\theta_d = \boxed{\theta_{d0}} + \boxed{c_d d + c_v v}$$

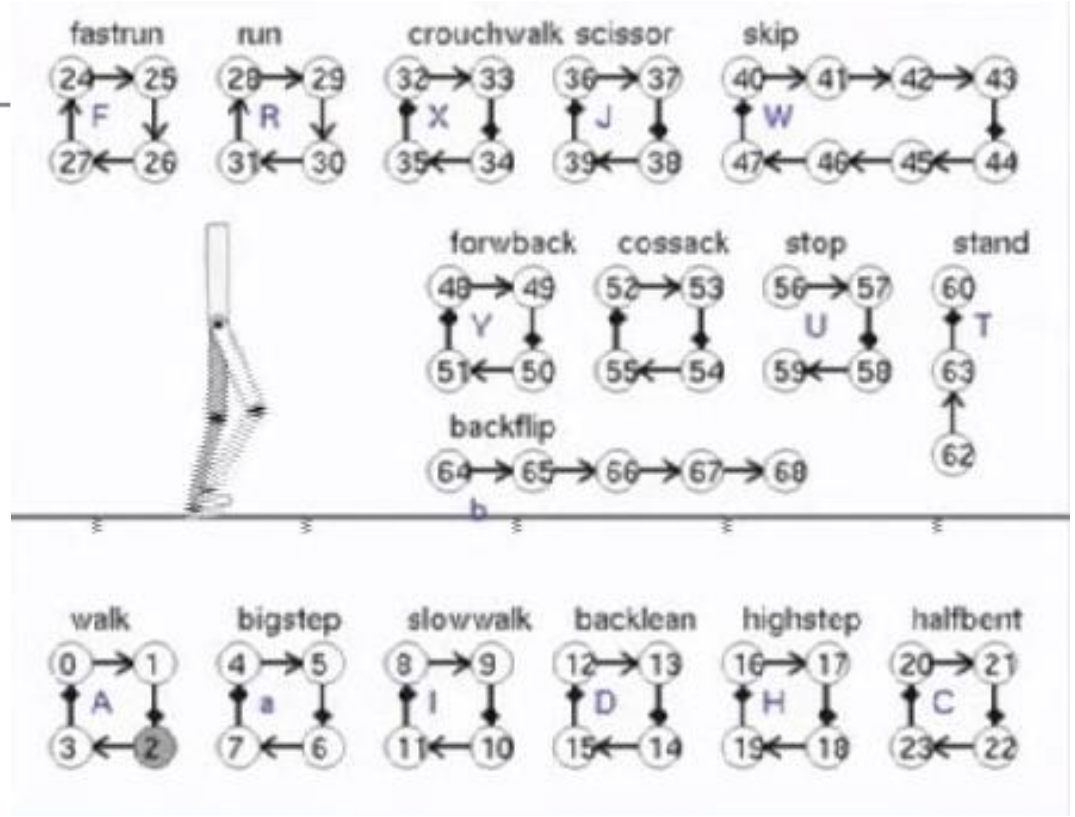
COM position

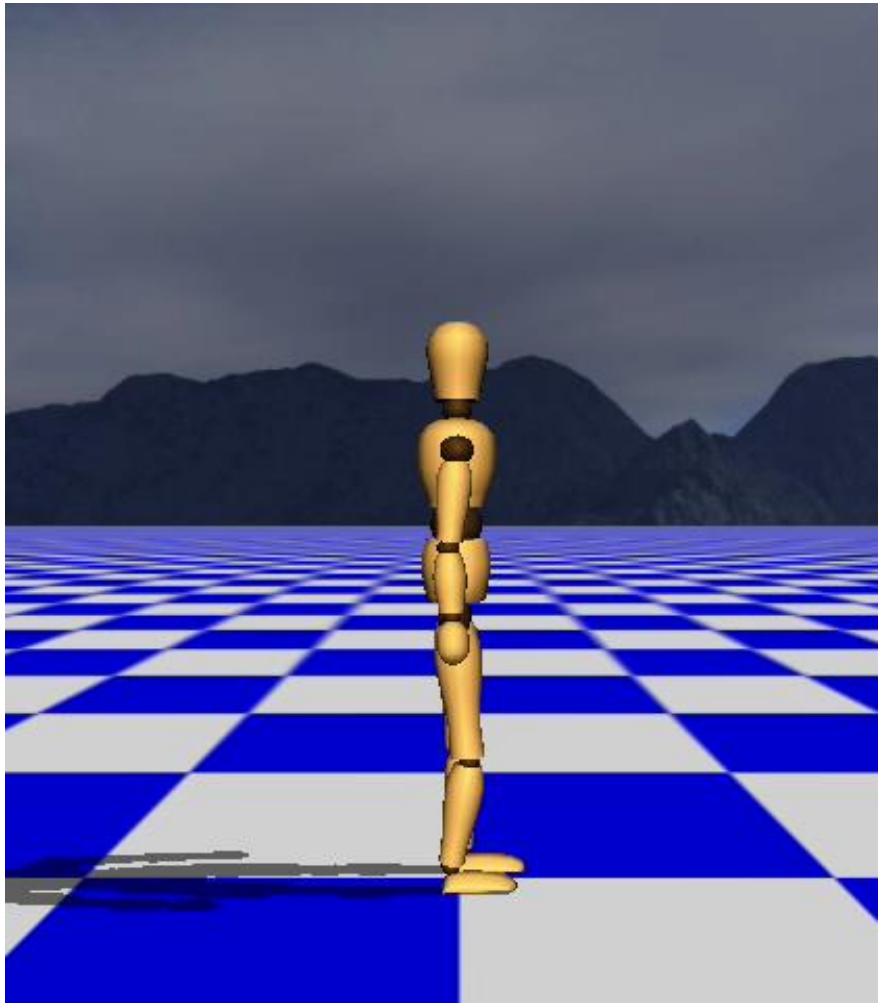
COM velocity

SIMBICON



Linear Feedback Control + FSM

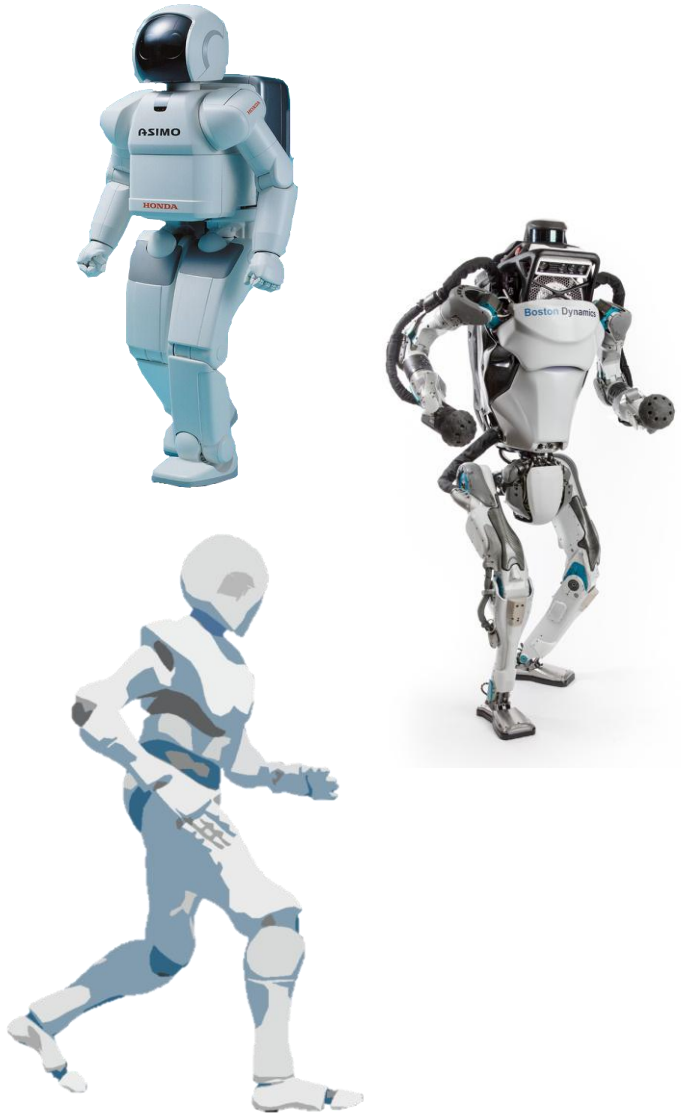




[Yin et al. 2007, SIMBICON]

Outline

- Walking and Dynamic Balance
- Simplified Models
  - ZMP (Zero-Moment Point)
  - Inverted Pendulum
  - SIMBICON
- How to generalize to other motion?



Questions?